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REVIEW

Spatiotemporal Pattern Mining of Moving Objects: Concepts, Methods, and Challenges

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Abstract
Mining potentially meaningful movement patterns of moving objects is of great significance for analyzing human movement law, understanding traffic mechanisms, and analyzing wild animal migration activities. In recent years, the growing availability of movement data from sensor networks and location-aware devices has provided great opportunities for investigating the spatiotemporal patterns of moving objects. A systematic review of this rapidly evolving field is necessary to synthesize existing knowledge and direct future research. In this paper, we systematically review the classification and mining methods of moving objects' movement patterns, identify eight major categories of movement patterns, and systematically compare existing pattern discovery methods, including classical spatial data mining techniques and emerging deep learning models. Based on this, we summarize the key challenges in current research, e.g., scale-adaptive pattern mining, real-time detection, co-mining of multiple patterns, and reproducibility. Furthermore, we outline promising future research directions, such as developing general pattern mining algorithms, integrating deep learning with traditional methods, and incorporating semantic context into pattern interpretation. This study can provide a reference for subsequent research and advance the further development of this field, especially promoting the expansion of current spatial data mining methods for static data to spatiotemporal patterns mining for dynamic streaming data oriented to moving objects.

Keywords Movement pattern · Moving object · Spatiotemporal data mining · Movement data

Introduction

Motivation

From microscopic subatomic particles to macroscopic celestial bodies, motion is ubiquitous and the most fundamental property of life (Nathan et al. 2008; Guibas and Roeloffzen 2017), involving spatial (e.g., location), temporal, and physical attribute (e.g., speed, direction) dimensions. Moving objects, broadly defined, encompass anything that changes its spatial position or state over time (Dodge et al. 2008). Moving objects can be tangible entities that move in geographical space, such as humans, animals, and vehicles, or abstract dynamic processes that propagate and evolve in space over time, such as hurricanes and infectious diseases (Li et al. 2023a). The

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activities carried out by these moving objects partly shape the dynamics and patterns of the world, from physical landscapes and ecosystems to social systems and sea-land-air operation networks. Consequently, understanding the movement process and mining patterns of moving objects is significant and has attracted extensive attention from experts and scholars in various domains and disciplines for decades.

Benefiting from the rapid development of positioning and sensing technology, massive data that track the movements of moving objects (i.e., movement data) have become increasingly available (Atluri et al. 2018; Graser et al. 2020). Movement data provides valuable spatiotemporal information that reveals both observed movement regularities and underlying behavioral intentions of moving objects. These rich movement data have promoted movement pattern analysis to emerge as a prominent research topic in the fields of geographical information science (GIS), computer science, data science, and behavioral science (Demšar et al. 2015, 2021; Miller et al. 2019). Movement pattern mining aims to reveal the distribution regularities and interactions of moving objects in space over time, which is helpful in a variety of real-world applications, such as group event discovery (Zheng et al. 2013), maritime management (Kontopoulos et al. 2021), behavior analysis (Renso et al. 2013), and intention inference (Yuan et al. 2014). Numerous studies have been conducted on movement representation, pattern mining and evaluation, visualization, and prediction. Dodge (2021) also established a movement data science paradigm, centered on visualizing, modeling, and analyzing movements to promote movement-related research.

In the Internet of Everything era, the availability of more fine-grained movement data has brought unprecedented opportunities and deepened the potential of movement research. With the rapid development of movement data collection and analysis techniques, it is necessary to conduct a systematic review of the current state of movement research. However, as an essential content in spatiotemporal data mining that has received much attention for a systematic survey, movement pattern mining lacks reviews that integrate the fragmented knowledge of existing movement research. This paper presents a comprehensive review of movement pattern research to bridge this gap. Specifically, the goal is to investigate the main concepts and techniques and to summarize common movement pattern types and pattern mining algorithms. In addition, we identify emerging trends and research gaps and address potential applications. We aim to provide a comprehensive landscape of the current movement pattern research development stage and offer valuable insights to researchers and others concerned to guide future advancements.

History

Even though movement pattern research is a relatively young field, it has received continuous attention and growing interest from diverse disciplines, especially regarding detecting various movement patterns for understanding moving objects' processes and underlying behaviors. Movement pattern mining and analysis is one of the most critical tasks in spatiotemporal data mining, which studies the moving dynamics of moving objects in space over time. The earliest study of movement originated from animal ecology (Devall 1980). Different movement behaviors of animals have been studied, such as migration patterns (Edelhoff et al. 2016), foraging (Bennison et al. 2018), and interactions with habitat (Heupel et al. 2019). Based on previous studies, researchers have proposed the framework of movement ecology (Holyoak et al. 2008) and the movement ecology paradigm for unifying movement research (Nathan et al. 2008). Later, the advances in movement data collection, geographic big data mining, and analysis further advanced ecological movement research (Kays et al. 2022; Nathan et al. 2022). Various spatial–temporal data mining methods (e.g., clustering, outlier detection) have been used to identify animal movement patterns and analyze animal behaviors.

In the field of GIS, Relative Motion (REMO), which focuses on the motion attributes of moving objects in relative space, is considered the foundation for the study of movement patterns (Laube and Imfeld 2002). Within the REMO framework, a motion pattern is a collection of moving objects and their motion attribute sequences that satisfy the predefined template. It distinguishes three basic motion patterns: sequences, incidences, and intersections. Further extensions incorporate spatial constraints and propose spatial motion patterns (Laube et al. 2005; Fisher et al. 2005). Thereafter, researchers have expanded the scope of movement pattern mining and analysis by improving existing movement patterns, defining new movement patterns, and developing various pattern mining algorithms and visual analytics techniques (Dodge et al. 2008; Andrienko and Andrienko 2013; Andrienko et al. 2021; Li 2014). Some classic movement patterns include moving clusters (Kalnis et al. 2005), flocks (Benkert et al. 2008), convoys (Jeung et al. 2008), swarms (Li et al. 2010a), gatherings (Zheng et al. 2013), snowball patterns (Guo et al. 2014), platoons (Li et al. 2015; Ma et al. 2021), and converging patterns (Zhao et al. 2020a; Jia et al. 2025). In addition to defining new patterns or expanding existing ones, scholars have also tried to establish systematic frameworks for classifying movement patterns. For example, Dodge et al. (2008) classify movement patterns into generic patterns (which are further divided into primitive patterns and compound patterns) and behavioral patterns with domain-specific semantics. Based on the number of moving objects involved, Li (2014) classified movement patterns into individual periodic patterns, pairwise trajectory patterns, and frequent trajectory patterns. Ji et al. (2016) divided movement patterns into two categories based on the directional consistency of moving objects: those moving in the same direction and those moving in opposite directions.

Movement pattern mining primarily focuses on identifying explicit distribution regularities in space over time exhibited by moving objects from a geometric perspective, without considering geographical context or semantic information. However, due to the generality of spatiotemporal representations, similar movement patterns may emerge across different types of moving objects (e.g., humans and animals) despite completely different behaviors or intentions (Dodge et al. 2009). To better interpret the underlying semantics (e.g., behaviors and intentions) of such movement patterns, it is necessary to incorporate additional semantic features (e.g., activity types, environmental factors, object-object metadata, and temporal context) into the movement pattern analysis (Baglioni et al. 2009; Vandecasteele et al. 2014; Peng et al. 2023). Semantic trajectory modeling and mining have become a key research direction in trajectory-related studies, focusing on enriching raw trajectories and extracting high-level behavioral patterns (Parent et al. 2013; Zhao et al. 2020a, b, c). Relevant studies have addressed tasks such as stop point detection (Zhao et al. 2021), trajectory segmentation (Gao et al. 2020; Liu et al. 2021), activity inference (Furletti et al. 2013), and semantic annotation (Yan et al. 2013). Building on semantically enriched trajectories, researchers have further explored semantic pattern mining approaches, including semantic similarity measures (Lehmann et al. 2019) and semantic trajectory clustering and classification (Liu and Guo 2020; May et al. 2020).

Recently, scholars in GIS have strived to develop an integrated science of movement, seeking to unify concepts and methods adaptable to various moving objects (Dodge et al. 2020; Demšar et al. 2021; Dodge 2021). Some studies also tried to develop unified approaches for discovering multiple movement patterns (Li et al. 2011; Wu et al. 2014; Francia et al. 2024). With the booming development of new technologies, researchers have also attempted to leverage artificial intelligence technology for movement pattern recognition. For example, Yang and Gidófalvi (2020) employed conventional neural networks to detect regionally dominant patterns. However, this field remains somewhat fragmented despite significant progress in movement pattern analysis. Therefore, there is an urgent need for an up-to-date, comprehensive review that summarizes existing studies and outlines future development trends to promote movement pattern mining and analysis.

Outline

The remainder of this article is as follows. The "Key Concepts and Basic Definitions" section introduces key concepts and definitions, and the "Moving Object Data: Characteristics" section analyzes the characteristics of movement data. The "Common Types of Movement Patterns" section classifies and describes classical movement patterns. The "Methodologies for Movement Pattern Mining" section reviews existing movement pattern mining methods. The "Applications of Movement Pattern Analysis" section presents typical applications, and the "Open Movement Datasets and Tools for Movement Pattern Mining" section discusses the challenges and future direction. The "Challenges and Future Directions" section concludes the study. The overall framework of the review is shown in Fig. 1.

Key Concepts and Basic Definitions

We first introduce the key concepts and basic definitions to establish a foundation for movement pattern research.

Definition 1 (*moving object*): Moving objects are those whose spatial locations or geometric states change over time. Moving objects can be visible, tangible entities (e.g., humans, vehicles, ships, and animals) and abstract objects or dynamic geographical phenomena that move over space and time (e.g., hurricanes and epidemic diseases).

Definition 2 (*movement*): Movement refers to the act or process of a moving object changing its spatial properties (e.g., position) or altering its motion attributes (e.g., speed, heading) over time.

Definition 3 (*movement data*): Movement data denotes digitally recorded observations of the movement of moving objects. It can be represented in various formats, such as discrete trajectories, continuous functions (e.g., parametric curves), event-based records, and origin–destination flows.

Definition 4 (*trajectory*): A trajectory is an ordered sequence of discrete points, each including the primitive spatiotemporal information (i.e., spatial coordinates and timestamps), as well as some other physical attributes (e.g., speed and direction). It is the most common form of movement data.

Definition 5 (*movement pattern*): A movement pattern (or spatiotemporal pattern) is defined as the discernible, non-random spatial and temporal arrangements or any interesting relationships recognized from movement data (Andrienko and Andrienko 2007; Dodge et al. 2008).

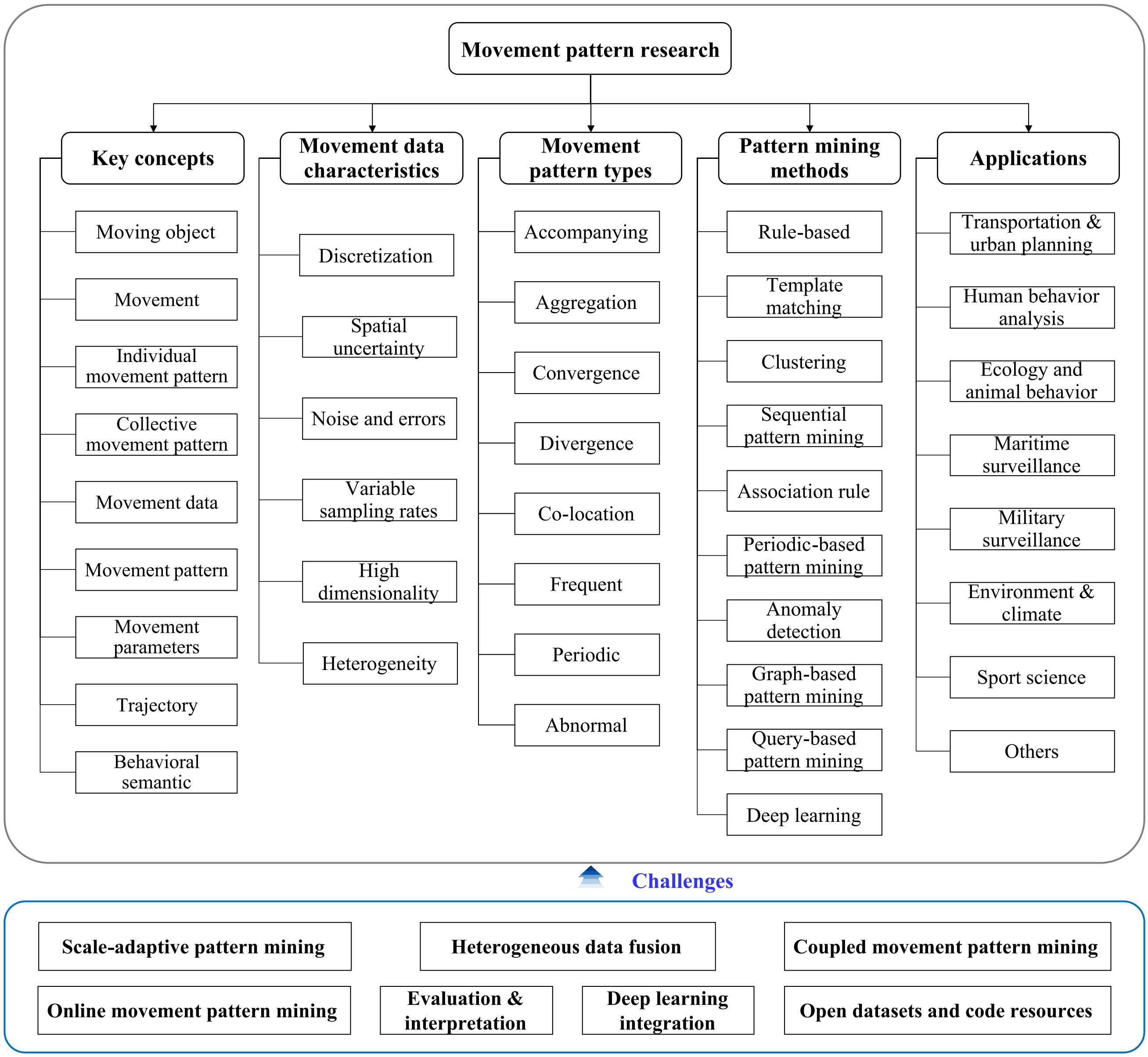


Fig. 1 The overall framework of the review

Definition 6 (*individual movement pattern*): Individual movement pattern denotes the pattern formed by only a single, independent moving object (Andrienko and Andrienko 2007).

Definition 7 (*collective movement pattern*): Collective movement pattern (or say group movement pattern) refers to the movement pattern formed by at least two moving objects (Dodge et al. 2008).

Definition 8 (*movement parameters*): Movement parameters refer to a series of quantitative attributes that characterize the spatiotemporal properties of moving objects. As illustrated in Dodge et al. (2008), movement parameters can be categorized into three levels: primitive parameters (e.g., spatial location, instance, and interval), primary derivatives (e.g., distance, direction, duration, and speed), and secondary derivatives (e.g., spatial distribution and temporal distribution).

Moving Object Data: Characteristics

Understanding the characteristics of moving object data (i.e., movement data) is crucial for developing effective movement analysis methodologies. While the movement of moving objects is inherently continuous, the limitations in data acquisition and representation introduce specific characteristics of movement data, which pose non-negligible challenges for movement pattern mining.

- Discretization: The movement process of moving objects is inherently continuous. However, their movements are typically recorded as sequences of discrete points, representing a discretized snapshot of this process. Understanding this discrete representation nature is critical for effective movement data modeling and analysis, and pattern mining. Although interpolation techniques such as kinematic and Lagrange interpolation (Long 2016; Wang et al. 2024) can be used to improve the granularity of movement data, they cannot completely reconstruct the underlying continuous movement paths.
- Uncertainty: The discrete sampling process, device system errors, inaccurate positioning, or environmental disturbances inevitably introduce potential information loss and uncertainty of movement data (Pfoser and Jensen 1999; Zheng et al. 2012). Such certainty refers to the inability to precisely determine the actual position and time of the moving object between consecutive recorded points. Besides spatial and temporal dimensions, attribute uncertainty (e.g., uncertain speed or direction) and contextual uncertainty (e.g., unknown environmental factors affecting the movement) may also exist and impact the mining and analysis of movement patterns.
- Variable sampling rates: Variations in data collection equipment or settings may result in different sampling rates or frequencies for different moving objects or different moving stages of the same moving object. This inconsistency makes it difficult to directly mine patterns from movement data and requires additional temporal alignment and spatial interpolation operations.
- Multidimensional features: Unlike simple numerical data, movement data includes not only basic spatial coordinates and time information, but also derived attributes such as speed, direction, and acceleration. This multifaceted information leads to the high dimensionality of movement data.
- Heterogeneity: Data collection devices and methods are different for different types of moving objects, leading to heterogeneity in data types, formats, and sampling rates (Dodge et al. 2016).

Common Types of Movement Patterns

Movement patterns can be classified into individual and collective movement patterns based on the number of moving objects involved. Generally, movement patterns observed in individuals are also present in groups. Therefore, we do not explicitly distinguish between individual and collective movement patterns. By summarizing existing research, we roughly categorize movement patterns into eight common types, as shown in Fig. 2. Each type of movement pattern is further subdivided into finer subtypes based on different constraints. Before introducing the movement patterns, we first define essential notations for mathematically formalizing movement patterns.

Let $\boldsymbol{O} = \{o_1, o_2, \ldots, o_n\}$ be the set of moving objects, $\boldsymbol{T} = \{T_1, T_2, \ldots, T_n\}$ be the set of historic movement tracks of moving objects, and $\boldsymbol{T}_{\mathbf{DB}} = \{t_1, t_2, \ldots, t_m\}$ be the set of all timestamps. Each track T_i is composed of m consecutive points, i.e., $T_i = \{p_{i1}(x_{i1}, y_{i1}, t_{i1}), \ldots, p_{im}(x_{im}, y_{im}, t_{im})\}$, where $1 \le i \le m$.

Accompanying Movement Patterns

Accompanying movement patterns describe a group of moving objects that travel together over a certain period. This category encompasses multiple derived patterns based on constraints, such as temporal continuity, group membership persistence, and spatial configuration. An overall comparative analysis of these patterns is presented in Table 2 (seen in the Appendix). Formal introductions are below.

Flock

The flock concept originates from animal collective behavior (e.g., bird flocks) (Okubo 1986). A *flock* is defined as at least m fixed moving objects traveling together within a disc of predefined size r for at least k consecutive timestamps (Gudmundsson and Van Kreveld 2006; Benkert et al. 2008; Vieira et al. 2009). Formally, $\boldsymbol{F}$ (m, k, r) is a flock that satisfies the following conditions:

- Spatial constraint: At any given time t, the moving objects in $\boldsymbol{F}$ are located within a radius r of each other, i.e., $\forall\, o_i, o_j \in F$, dist $(p(o_i, t), p(o_j, t)) \le 2r$.
- Temporal continuity: The moving objects in $\boldsymbol{F}$ must maintain the spatial constraint for at least k consecutive timestamps.
- Minimum size: The flock must contain at least m moving objects.
- Membership stability: The moving objects in $\boldsymbol{F}$ remain unchanged over the duration of k.

Based on the stability of group membership, Gudmundsson and Van Kreveld (2006) distinguished two flock variants: *fixed flocks* and *varying flocks*. Depending on whether the spatial location of the defined disk is fixed (or exhibits little change) or variable, flocks can be categorized as *stationary flocks* (also say *meet*) or *moving flocks* (Wachowicz et al. 2011; Turdukulov et al. 2014; Zhang and Van de

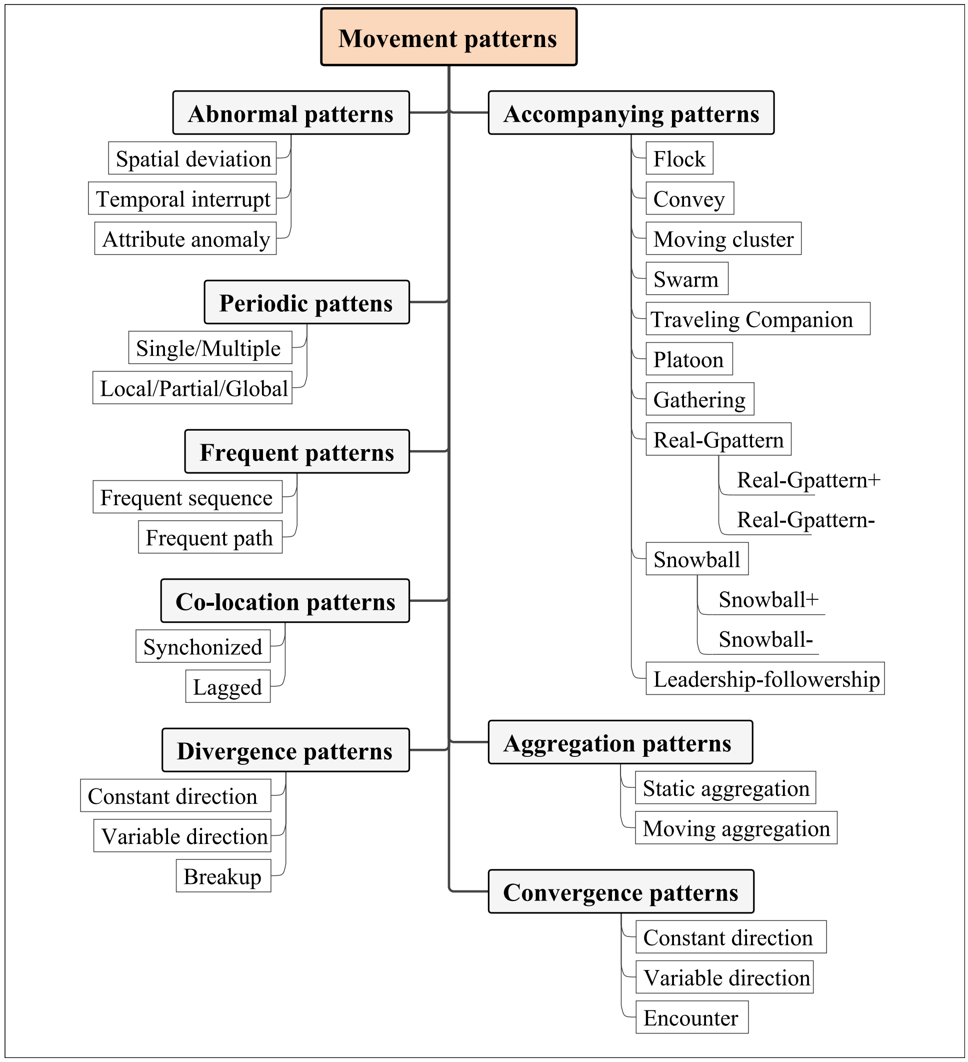


Fig. 2 Eight common types of movement patterns and their variants

Weghe 2024), as illustrated in Fig. 3. However, due to its circular constraint of spatial shape, the flock pattern cannot reveal natural groups with irregular shapes, leading to the lossy-flock problem (Jeung et al. 2008).

Convoy

A *convoy* is defined as a group of at least m fixed moving objects that are density-connected for at least k consecutive

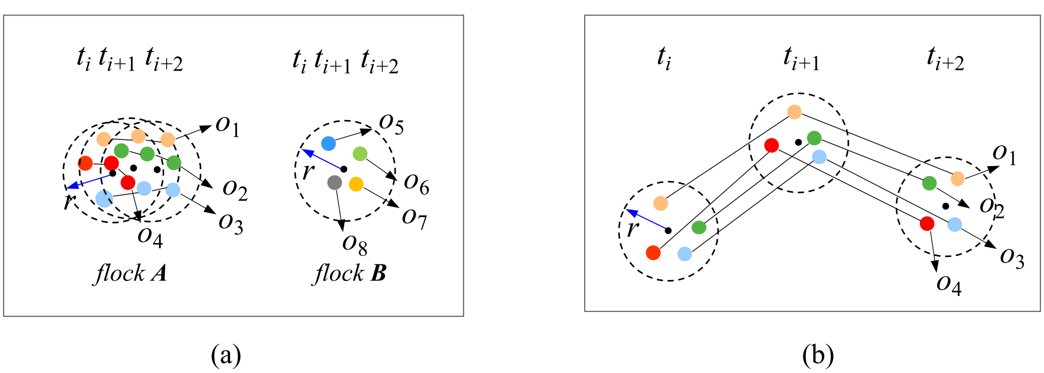


Fig. 3 Illustration of the flock pattern. **a** Two stationary flocks (left: only slight movement variation, right: completely stationary) and **b** a moving flock. The parameter r represents the radius of the disc used to define the spatial proximity of moving objects

timestamps, which overcomes the lossy-flock problem inherent in the flock pattern (Jeung et al. 2008; Yeoman and Duckham 2016; Yadamjav et al. 2020; Liu et al. 2023). The concept of density connection originates from the density-based clustering algorithm, and more details can be found in the literature by Ester et al. (1996). Formally, a convoy $\boldsymbol{C}$ (m, k, ε) is a subset of moving objects $\boldsymbol{O}$ that satisfy the following conditions:

- Density-connected: At any given time t, the moving objects in $\boldsymbol{C}$ are density-connected within a neighborhood of radius ε, i.e., $\forall o_i, o_j \in \boldsymbol{C}, \exists o_1, o_2, \ldots, o_q \in \boldsymbol{C}$ such that $o_1 = o_i$, $o_q = o_j$, and dist $(o_p, o_{p+1}) \le \varepsilon$ for $1 \le p < q$.
- Temporal continuity: The moving objects in $\boldsymbol{C}$ must maintain the density-connected constraint for at least k consecutive timestamps.
- Minimum size: The convoy must contain at least m moving objects.
- Membership stability: The moving objects in $\boldsymbol{C}$ remain unchanged over the duration of k.

The standard convoy pattern requires a fixed set of moving objects, where no members join or leave during the pattern. To relax this constraint, two variants, *evolving convoys* and *dynamic convoys*, have been proposed (Aung and Tan 2010), as shown in Fig. 4c and d. The evolving convoy allows for a gradual increase or decrease in the number of pattern members. In contrast, the dynamic convoy permits its members to be briefly absent without breaking the continuity of the convoy.

Moving Cluster

A *moving cluster* is defined as a group of moving objects that are density-connected during k consecutive timestamps, which relaxes the strict membership requirement of flock and convoy patterns (Kalnis et al. 2005; Yang et al. 2022). It only requires the Jaccard Index between the set of moving objects at consecutive timestamps to be no less than a given threshold θ. An example of a moving object is shown in Fig. 5. Let $\boldsymbol{g} = \{c_1, c_2, \ldots, c_k\}$be a sequence of snapshot clusters; then, $\boldsymbol{g}$ (ε, k, θ) is a moving cluster that satisfies the following conditions:

- Density-connected: At any given time t, the moving objects in the snapshot cluster c_i $(1 \le i \le k)$ are density-connected within a neighborhood of radius ε, i.e., $\forall o_i, o_j \in \boldsymbol{C}, \exists o_1, o_2, \ldots, o_m \in \boldsymbol{g}$ such that $o_1 = o_i$, $o_q = o_j$, and dist $(o_p, o_{p+1}) \le \varepsilon$ for $1 \le p < q$.
- Jaccard constraint: For any two consecutive clusters c_t and c_{t+1}, the Jaccard Index of their moving objects must exceed a given threshold θ $(0 < \theta \le 1)$, i.e., $|c_t \cap c_{t+1}| / |c_t \cup c_{t+1}| \ge \theta$.
- Minimum temporal duration: It must persist for at least k consecutive timestamps.

Swarm

A *swarm* is defined as a group of at least m fixed moving objects that are density-connected for at least k timestamps, which can be non-consecutive (Li et al. 2010a; Qian et al. 2022). As presented in Fig. 6a, the swarm relaxes the

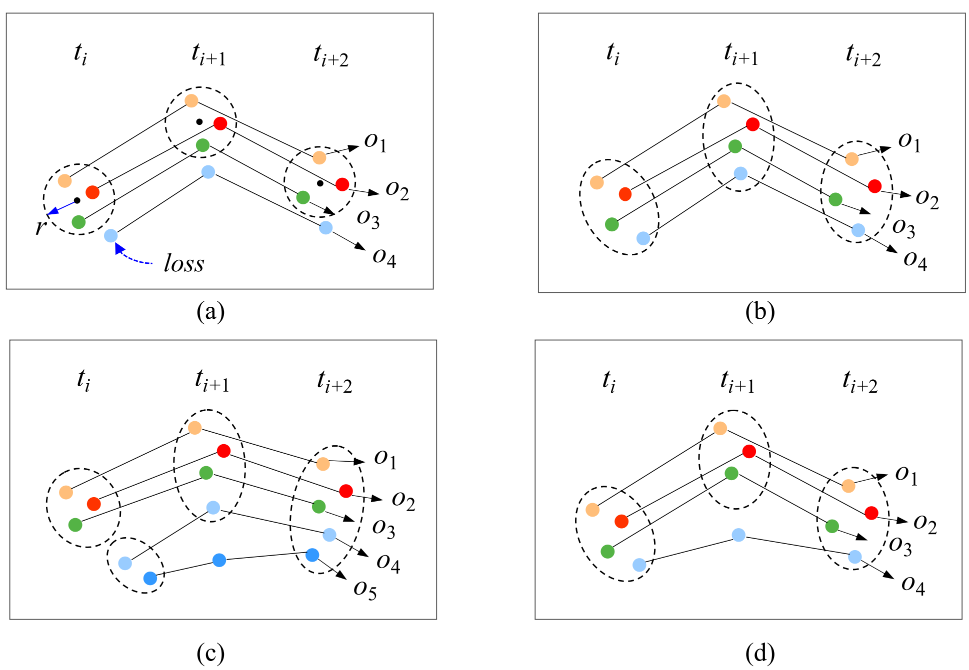
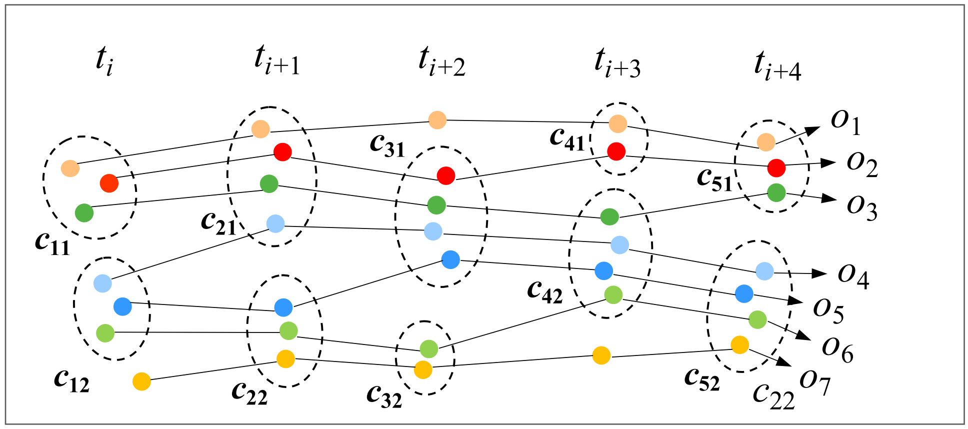


Fig. 4 Illustration of different convoy patterns. **a** The lossy-flock problem (o_4 is lost); **b** a standard convoy with stable membership across all timestamps (o_1, o_2, o_3, and o_4); **c** an evolving convoy with objects joining the group over time (o_4 joins at t_{i+1} and o_5 joins at t_{i+2}); **d** a dynamic convoy that allows temporal absence of objects (o_4 is absent at t_{i+1})

Fig. 5 Illustration of the moving cluster pattern where $c_{11},\ldots, c_{52}$ are snapshot clusters. Given parameters $k=4$ (minimum consecutive timestamps), $\theta=2/3$ (minimum moving object overlap ratio between consecutive timestamps), the snapshot cluster sequence $<c_{11}, c_{21}, c_{31}, c_{43}, c_{52}>$ forms a valid moving cluster



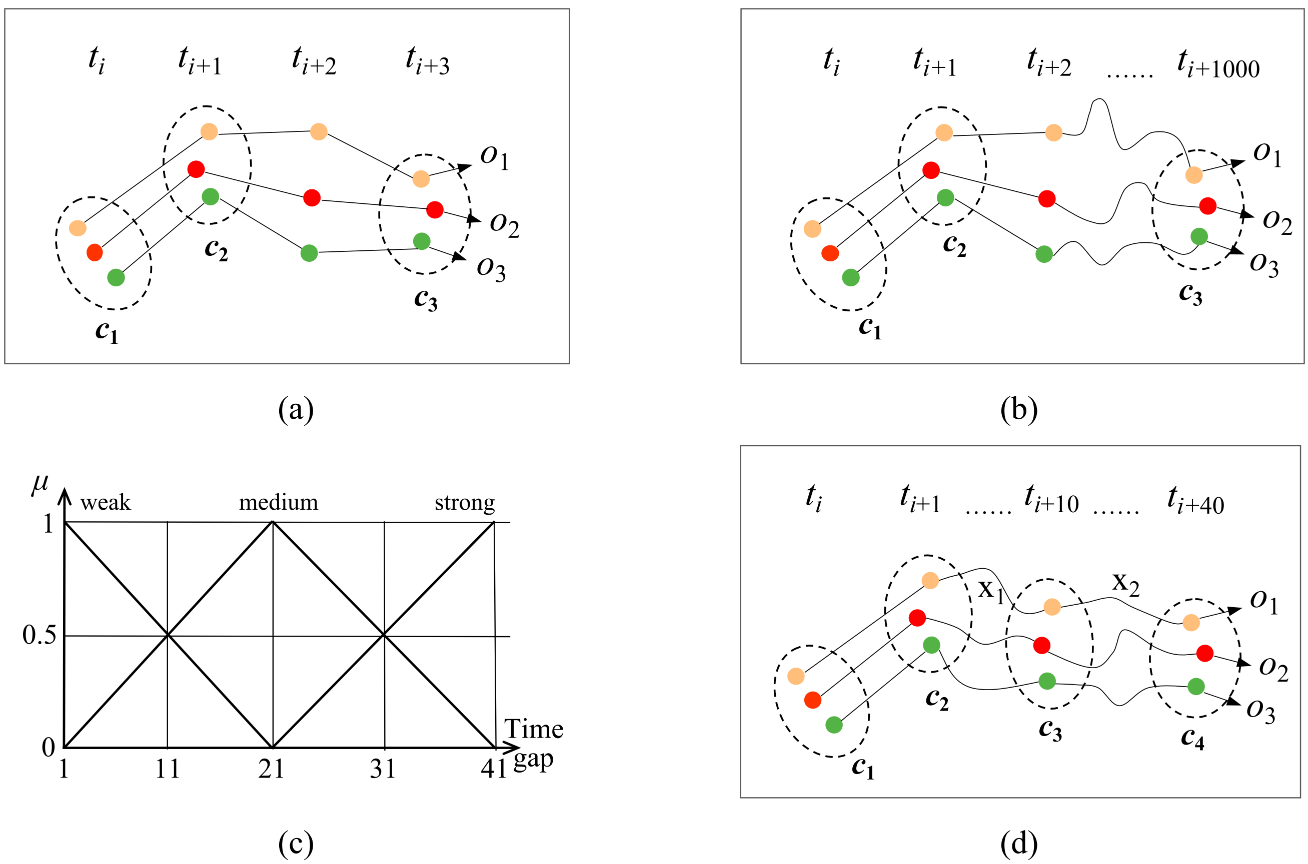


Fig. 6 Illustration of the swarm pattern. **a** A standard swarm, where objects o_1, o_2, and o_3 are not co-located at t_{i+2}, **b** a swarm with prolonged time gaps between co-occurrences, **c** illustration of a membership degree function used to quantify the strength of temporal gaps (Hai et al. 2012), and **d** a fuzzy swarm where time gaps x_1 and x_2 are mapped to a fuzzy set including weak, medium, and strong levels

temporal continuity requirement of the convoy pattern. Formally, given a neighborhood threshold ε, $\boldsymbol{S}$ (m, k, ε) is a swarm that satisfies the following conditions:

- Density-connected: Same as the convoy pattern.
- Minimum temporal duration: The moving objects in $\boldsymbol{S}$ must persist for at least k timestamps.
- Minimum size: The swarm must contain at least m moving objects.
- Membership stability: The moving objects in $\boldsymbol{S}$ remain unchanged over the duration of k.

The standard swarm pattern completely relaxes the temporal continuity constraint (i.e., the time gaps can be arbitrarily long), which may generate extraneous and meaningless patterns, as illustrated in Fig. 6b. To address this issue, Hai et al. (2012a) proposed the fuzzy swarm, which softens the temporal gap constraint by incorporating membership degree functions from fuzzy set theory.

Traveling Companion

A traveling companion is defined as a group of moving objects that move together for a certain period. A novel data structure called *traveling buddy* has been designed to identify and maintain micro-groups of closely related objects, making it efficient for online traveling companion pattern mining (Tang et al. 2012; Tang et al. 2014; Chen et al. 2016; Puntheeranurak et al. 2018; Yao et al. 2020). An illustration of the traveling companion pattern and traveling buddy structure is shown in Fig. 7. Let δ_γ be the buddy radius threshold, δ_t be the minimum duration threshold, and $O(t)$ be the moving objects at time t. A traveling buddy $\boldsymbol{b}$ $(\delta_\gamma, \delta_t)$ is defined as a set of moving objects that satisfy: (1) $\boldsymbol{b} \in O(t)$, and (2) for $\forall o_i \in \boldsymbol{b}$, dist $(o_i, \mathrm{cen}(\boldsymbol{b})) \le \delta_\gamma$, where cen($\boldsymbol{b}$) denotes the geometric center of $\boldsymbol{b}$. The radius of the traveling buddy, γ, is defined as the maximum distance from cen($\boldsymbol{b}$) to any of its members. The variants of the traveling companion pattern include loose traveling companions that relax the time continuity constraint (Naserian et al. 2018), loose group companions, and evolving companions that allow some members to be absent at specific timestamps (Shein et al. 2020a, b).

Platoon

A *platoon* is defined as a group of moving objects that travel together for some consecutive time segments, which allows time gaps (i.e., locally consecutive) (Li et al. 2015; Ma et al. 2021). Formally, let $\boldsymbol{g} = \{c_1, c_2, \ldots, c_k\}$ be a sequence of snapshot clusters. Given a minimum number of moving objects min_o, a minimum number of timestamps min_t, a minimum number of consecutive timestamps min_c, and a neighborhood radius ε, $\boldsymbol{P}(\varepsilon, min_o, min_t, min_c)$ is a platoon that satisfies the following conditions:

- Density-connected: Same as moving clusters.
- Minimum temporal duration: The platoon $\boldsymbol{P}$ must persist for at least min_t timestamps.
- Local continuity: The platoon $\boldsymbol{P}$ must persist for at least min_c consecutive timestamps.
- Minimum size (significantsupport): The platoon $\boldsymbol{P}$ must contain at least min_o moving objects.

Figure 8 shows an example of a platoon pattern. The main difference between the platoon pattern and the accompanying patterns introduced above is their temporal constraint. The platoon pattern allows temporal gaps but still requires local temporal continuity. This balances the need to capture the cohesive motion of moving objects

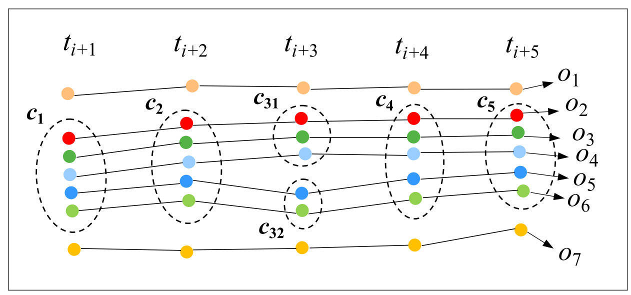


Fig. 8 Illustration of the platoon pattern, where $c_1, \ldots, c_5$ are snapshot clusters. Moving objects o_2, o_3, o_4, o_5, and o_6 travel together as a platoon at timestamps t_{i+1}, t_{i+2}, and t_{i+5}

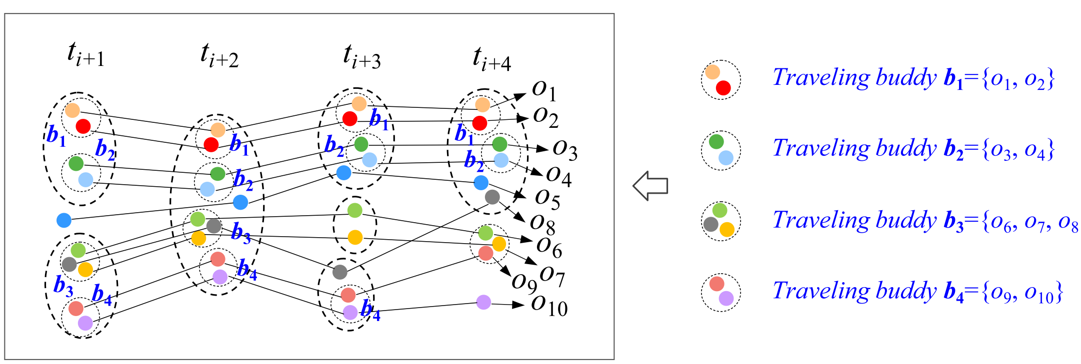


Fig. 7 Illustration of the traveling companion pattern with traveling buddies (Tang et al. 2012). At t_{i+1}, four buddies are initialized as two traveling companions: $r_1 = \{b_1, b_2\}$ and $r_2 = \{b_3, b_4\}$; at t_{i+2}, four buddies remain, with $r_1 = \{b_1, b_2\}$, $r_2 = \{b_3, b_4\}$, and a new traveling companion $r_3 = \{ b_1, b_2, b_3, b_4, o_5\}$; at t_{i+3}, b_3 disappears while b_1, b_2, b_4 persist; comparisons are updated to $r_1 = \{b_1, b_2\}$, $r_2 = \{b_4, o_8\}$, $r_3 = \{ b_1, b_2, o_5\}$; at t_{i+4}, b_4 disappears, and the remaining companions are $r_1 = \{b_1, b_2\}$, $r_3 = \{ b_1, b_2, o_5\}$. Ultimately, $\{o_1, o_2, o_3, o_4\}$ is identified as a persistent traveling companion from t_{i+1} to t_{i+4}

within consecutive intervals while accommodating the intermittent disruptions that may occur in the real world.

Gathering

A *gathering* is defined as a set of at least m_c moving objects that move together continuously for at least k_c consecutive timestamps and maintain at least m_p moving objects throughout the pattern (Zheng et al. 2013; Zhang et al. 2014; He et al. 2020). Formally, given a neighborhood radius threshold ε and a cluster distance threshold δ, let $\boldsymbol{g} = \{c_1, c_2, \ldots, c_k\}$ be a sequence of snapshot clusters. Then, $\boldsymbol{G}$ $(m_c, m_p, k_c, \varepsilon, \delta)$ is a gathering pattern that satisfies the following conditions:

- Density-connected: Same as the convoy pattern.
- Snapshot cluster distance constraint: The distance (e.g., Hausdorff distance) between any two consecutive snapshot clusters does not exceed δ, i.e., $\forall\ ct_i, ct_{i+1} \in C_t$, $\text{dist}(ct_i, ct_{i+1}) \leq \delta$.
- Minimum gathering support constraint: At any given time t, the number of moving objects in $\boldsymbol{G}$ (or, equivalently, the snapshot cluster c_t) must be no less than the specified minimum support size m_c, i.e., $|c_t| \geq m_c$, where $|c_t|$ denotes the number of moving objects in c_t.
- Temporal duration constraint: The lifetime of $\boldsymbol{G}$ must last for at least k_c timestamps.
- Persistent participatorconstraint: At least m_p participants in each snapshot cluster of $\boldsymbol{G}$, where a moving object is a participator if it appears in at least k_p snapshot clusters.

As illustrated in Fig. 9, the gathering pattern addresses the large congregations of moving objects with stable and durable group members while accommodating dynamic interactions within the group, such as the joining and leaving of peripheral members.

Real-Gpattern

A *real-Gpattern* is defined as a pattern where the number of moving objects that travel together gradually increases or decreases for at least k timestamps (Hai et al. 2012b; Zhang et al. 2019). Formally, let $\boldsymbol{g} = \{c_1, c_2, \ldots, ck\}$ be a sequence of snapshot clusters. Then, $\boldsymbol{G}^*(k)$ is a real-Gpattern that satisfies the following conditions:

- Minimum temporal duration: The $\boldsymbol{G}^*$ pattern must persist for at least k timestamps (temporal continuity is unnecessary), meaning it must contain at least k snapshot clusters.
- Graduality constraint: The $\boldsymbol{G}^*$ pattern must exhibit a gradual evolution in the number of participating moving objects over time, which includes two situations: (a) Gradual increase: Each snapshot cluster contains at least all moving objects from the preceding snapshot cluster, with the final cluster containing strictly more objects than the initial cluster, i.e., $\forall\ i\ (1 \leq i < k)$, $c_i \subseteq c_{i+1}$, and $|c_k| > |c_1|$; (b) gradual decrease: Each snapshot cluster contains at most all moving objects from the preceding snapshot cluster, with the final cluster containing strictly fewer objects than the initial cluster, i.e., $\forall\ i\ (1 \leq i < k)$, $c_i \supseteq c_{i+1}$, and $|c_k| < |c_1|$.

An example of the *real-Gpattern* is illustrated in Fig. 10. This pattern can be considered either an accompanying movement pattern, where a group of gradually increasing or decreasing moving objects travels together, or an aggregation movement pattern, introduced next, where an increasing number of objects gather and move towards a similar direction.

Snowball

A *snowball* is defined as a cluster of moving objects that either rapidly attracts nearby moving objects (growing)

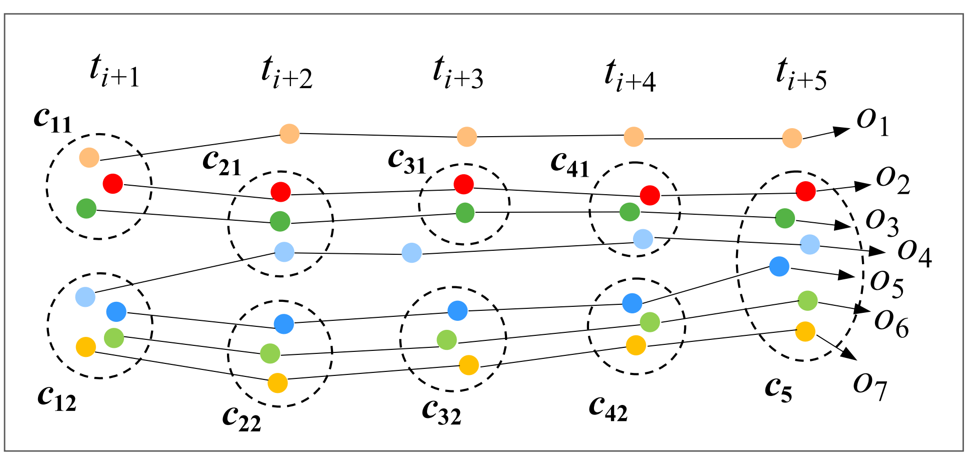


Fig. 9 Illustration of the gathering pattern. Given parameters $m_c = 2$, $m_p = 3$, $k = 4$, the sequence $< c_{12}, c_{22}, c_{32}, c_{42}, c_5 >$ forms a valid gathering pattern. It includes three participants (o_5, o_6, o_7) who are present from t_{i+1} to t_{i+5} (five timestamps)

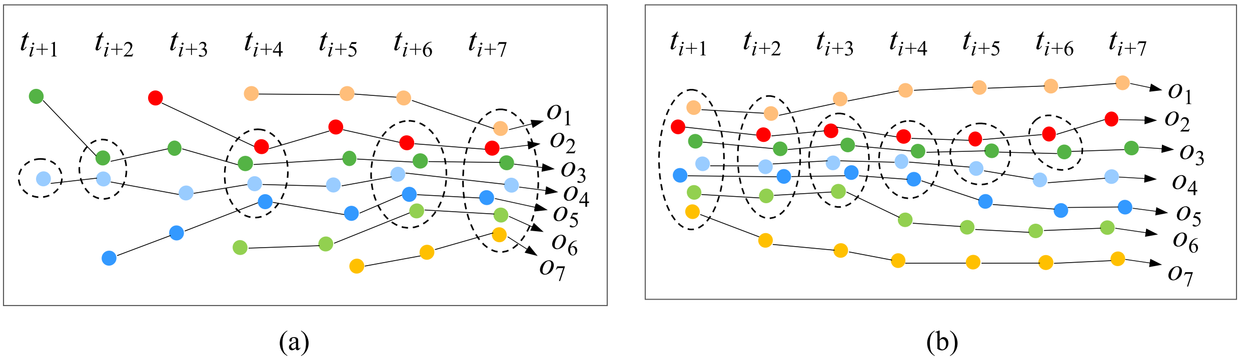


Fig. 10 Illustration of the real-Gpattern. **a** A real-Gpattern+, where the number of participating moving objects gradually increases over time with a temporal gap occurring at t_{i+5}, and **b** a real-Gpattern-, where the number of participating moving objects gradually decreases

or sheds its members (shrinking), resulting in significant size changes over at least δ_t consecutive timestamps (Guo et al. 2014). Formally, let $\boldsymbol{g} = \{c_{t1}, c_{t2}, \ldots, c_{tn}\}$ be a sequence of clusters at consecutive timestamps. Given an integrity threshold δ_θ and a speed threshold δ_{speed}, $\boldsymbol{SN}(\delta_\theta, \delta_{speed}, \delta_t)$ is a snowball pattern that satisfies the following conditions:

- Cluster-connected: Any two consecutive clusters, c_{ti} and c_{ti+1} are connected if they share a sufficient number of common moving objects and the size change rate meets the predefined threshold. For a snowball + pattern, $\forall c_{ti+1} \in \boldsymbol{SN}$ $(t_1 \leq t_i < t_n)$ is cluster-connected + to $c_{ti} \in \boldsymbol{SN}$ if: $|c_{ti} \cap c_{ti+1}|/|c_{ti}| \geq \delta_\theta$ and $(|c_{ti+1}|\text{-}|c_{ti}|)/|c_{ti}| \geq \delta_{speed}$. For a snowball- pattern, $\forall c_{ti+1} \in \boldsymbol{SN}$ $(t_1 \leq t_i < t_n)$ is cluster-connected- to $c_{ti} \in \boldsymbol{SN}$ if $|c_{ti} \cap c_{ti+1}|/|\ c_{ti+1}| \geq \delta_\theta$ and $(|c_{ti}|\text{-}|c_{ti+1}|)/||c_{ti+1}|| \geq \delta_{speed}$.
- Temporal duration: The formation of the snowball pattern $\boldsymbol{SN}$ must span at least δ_t consecutive timestamps, i.e., $t_n\text{-}t_1 \geq \delta_t$.

As previously described, moving clusters and gatherings are the two patterns most closely related to the snowball pattern, which integrates their core ideas while adding constraints on growth or ablation rates. Figure 11 illustrates examples of snowball + and snowball- patterns.

Leadership-Followership

A *leadership-followership* pattern describes a group of moving objects (referred to as *followers*) that accompany another moving object (referred to as the *leader*) over time, where the movements of followers are spatially and directionally influenced by the leader (Andersson et al. 2007, 2008). This pattern can be considered a type of accompanying movement pattern, as the leader and followers move together for a sustained period. Formally, given a minimum number of followers m, a minimum duration of interaction k, a spatial radius r defining the extent of the awareness of followers,

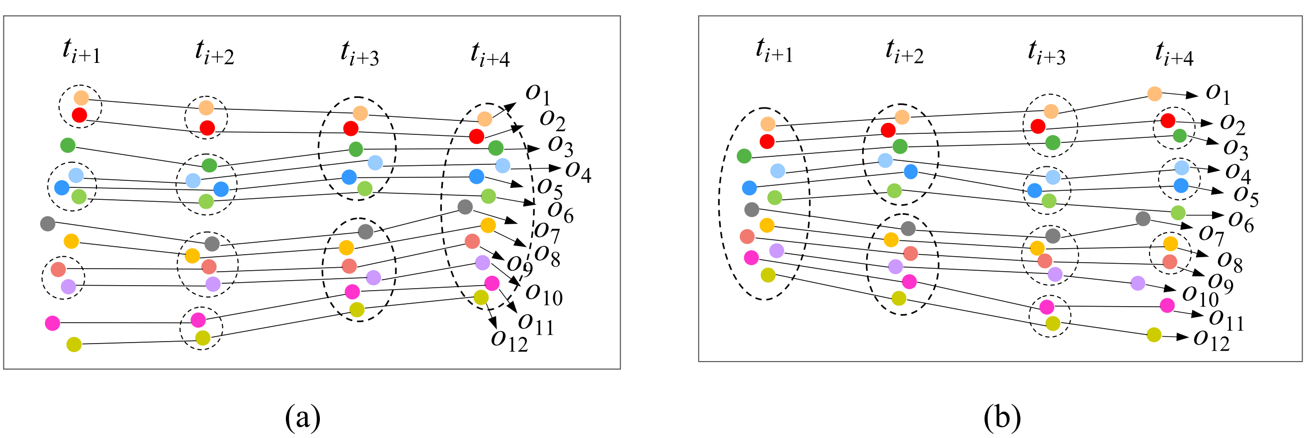


Fig. 11 Illustration of the snowball pattern. **a** Snowball +, where the number of participating moving objects increases progressively over time, and **b** snowball-, where the number of participating moving objects decreases progressively over time

an angle α defining the follower's forward-looking field of view, and a maximum angular deviation β between the directions of the leader and followers. Let $\boldsymbol{d} = \{d_1, d_2, \ldots, d_k\}$ be the sequence of objects' movement directions. Then, $\boldsymbol{LF}$ (m, k, r, β, α) is a leadership-followership pattern that satisfies the following constraints:

- Spatial constraint: At each timestamp t during the consecutive interaction duration k, for each follower o_f, the leader o_l must lie within its front region, defined as a sector with radius r and angle α centered on the follower's movement direction. And no non-following objects appear within the leader's front region, i.e., $\forall o_f \in \boldsymbol{LF}$: $o_l \in \text{front}(o_f) \wedge \forall o_j \in \boldsymbol{O} \backslash \boldsymbol{LF}$: $o_j \notin \text{front}(o_l)$.
- Directionalconstraint: At each timestamp t, the angular difference between the movement direction of each follower o_f and that of the leader o_l must not exceed the angle threshold β.
- Temporal persistence: The $\boldsymbol{LF}$ pattern must persist for at least k consecutive timestamps.
- Minimum follower count: The number of followers adhering to the spatial, directional, and temporal constraints must be greater than or equal to the minimum follower count m.

Figure 12 illustrates the leadership-followership pattern and the front region. This pattern often occurs in contexts such as animal migration, group foraging, guided tours, military formations, and coordinated vehicle fleets, where individuals typically rely on a leader for guidance or coordination.

Aggregation Movement Patterns

Aggregation movement patterns describe the spatial concentration of multiple moving objects (Dodge et al. 2008). Unlike accompanying movement patterns that emphasize the process of moving together, this type of movement pattern does not particularly concern how moving objects move before forming an aggregation but focuses on the final aggregated state. The spatial arrangement observed at a specific timestamp within an accompanying pattern (e.g., a flock) can be considered an instance of instantaneous spatial concentrations. Based on the nature of the aggregated area, we categorize the aggregation movement pattern into static aggregation and moving aggregation.

Static Aggregation

A *static aggregation* (or concentration) occurs when moving objects gather within a fixed area and remain stationary or hover, similar to the *stationary flock* and *static meet* pattern. Practical examples include traffic congestion at intersections, crowd gatherings at significant events, and queues forming at ticket counters. Formally, a static aggregation pattern is defined as a set of moving objects $\boldsymbol{SA} \in \boldsymbol{O}$ that satisfy the following conditions:

- Spatial proximity: At any given time t, the moving objects in $\boldsymbol{SA}$ must be spatially close to each other. This proximity can be measured using methods such as density connection (as in density-based clustering) or a predefined distance threshold.
- Temporal persistence: The moving objects in $\boldsymbol{SA}$ must remain in spatial proximity for at least k timestamps.
- Minimum size: There must be at least min_o moving objects in $\boldsymbol{SA}$ at any given time t.
- Stationarity: The spatial extent of the aggregated area remains relatively stable or confined to a specific region, which can be measured by ensuring that the Hausdorff or centroid distance between moving objects at consecutive timestamps is below a predefined threshold.

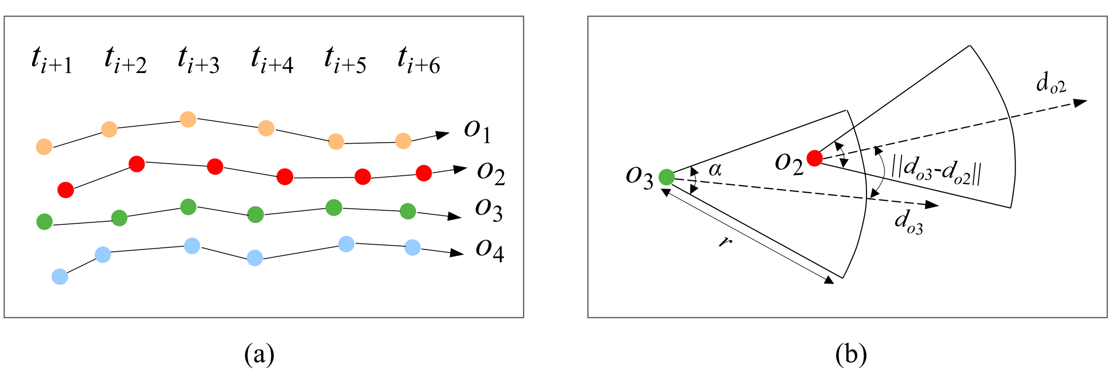


Fig. 12 Illustration of the leadership-followership pattern. **a** A leadership-followership pattern with four moving objects, where o_2 is the leader, and o_1, o_3, and o_4 are followers, and **b** an illustration of the front region (Andersson et al. 2007)

Moving Aggregation

A *moving aggregation* (or say varying meet) describes a situation where the aggregation area changes dynamically over time (Alamri et al. 2014; Dodge et al. 2008). After forming, the aggregated moving objects (or say a cluster) continue to move together. It allows surrounding objects to join or exit the cluster during movement, causing the cluster size to vary. Specific accompanying movement patterns can also be viewed as instances of moving aggregation, especially the gathering, real-Gpattern, and snowball patterns, as they capture the progressive growth and dynamic evolution of group members traveling together. Typical examples of moving aggregations include a herd of wildebeest migrating southward and a flock of birds maintaining cohesive formations during seasonal journeys. Formally, a moving aggregation pattern is defined as a set of moving objects $\boldsymbol{MA} \in \boldsymbol{O}$ that satisfy the following spatial range constraint while maintaining the same *spatial proximity*, *temporal persistence*, and *minimum size* constraints as the static aggregation pattern.

- Spatial range constraint: The spatial range of the aggregated area changes over time. This change can be quantified by metrics such as the Hausdorff or centroid distance between the moving object sets at consecutive timestamps, exceeding a predefined distance threshold.

Convergence Movement Patterns

The convergence movement pattern models spatiotemporal processes where moving objects from diverse locations gradually converge within a specific spatial region, forming a concentrated group from initially dispersed entities (Dodge et al. 2008; Fang et al. 2017). Early research, particularly in wildlife studies, defined convergence patterns as requiring moving objects to largely maintain their initial movement directions while allowing asynchronous arrival times at the target area (Laube et al. 2005), as illustrated in Fig. 13a. Under this context, a convergence pattern is defined as a set of moving objects $\boldsymbol{CP} \in \boldsymbol{O}$ that satisfy the following conditions:

- Direction preservation: At any timestamp t, the movement direction of each object $o_i \in \boldsymbol{CP}$ remains approximately unchanged from its initial direction.
- Spatial convergence constraint: The motion azimuth vectors of any two objects must intersect at a point p located within a predefined circular region $\boldsymbol{R}$, a circle of radius r centered at point c, that is, $\forall o_i, o_j \in \boldsymbol{CP}$, $\exists\, p$ such that $p \in \mathrm{Line}(p_i(t), v_i) \cap \mathrm{Line}(p_j(t), v_j)$ and $\mathrm{dist}(p, c) \le r$, where $p_i(t)$ and $p_j(t)$ are the positions of o_i and o_j at time t, v_i and v_j are their respective azimuth vectors, and $\mathrm{Line}(p_i(t), v_i)$ and $\mathrm{Line}(p_j(t))$ represent the lines extending from $p_i(t)$ in the direction of v_i and $p_j(t)$ in the direction of v_j, respectively.
- Asynchronous arrival: Moving objects in $\boldsymbol{CP}$ are allowed to reach region R at different times.

However, this definition of convergence presents an idealized scenario, assuming that moving objects maintain constant directions throughout their movements. In reality, objects often adjust their directions in response to

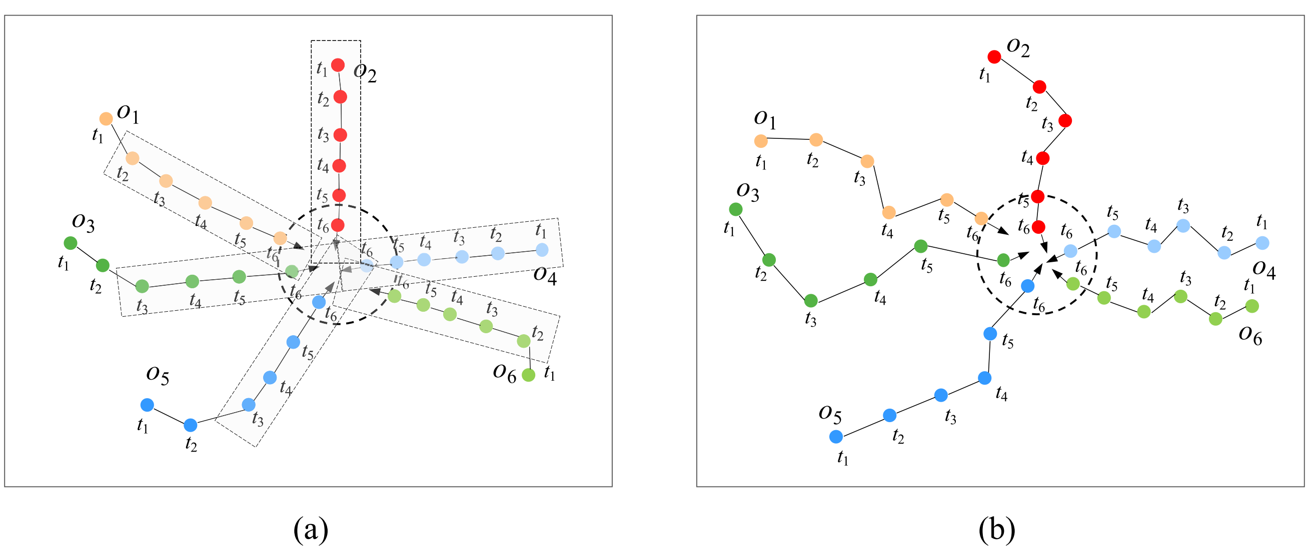


Fig. 13 Illustration of the convergence movement pattern. **a** Moving objects o_1, o_2, o_3, o_4, o_5, and o_6 maintain largely constant directions during the converging process, and **b** moving objects o_1, o_2, o_3, o_4, o_5, and o_6 converge from different directions, showing noticeable directional deviations during their movements

environmental factors, obstacles, or individual behaviors. For instance, individuals traveling from various locations to a stadium may alter their paths to avoid congestion, and foraging ants may dynamically adjust their routes based on pheromone trails. To better reflect real-world complexities, the direction constraint can be relaxed, allowing for deviations in the movement directions or only requiring that moving objects originate from different directions (Fig. 13b), provided they ultimately converge within the predefined region (Zhao et al. 2020a; Jia et al. 2021, 2025). Encounter (Dodge et al. 2008; Chen et al. 2023) is a variant of the convergence pattern that requires moving objects to arrive at a specific region synchronously.

Divergence Movement Patterns

The divergence movement pattern models the inverse spatiotemporal process of convergence, where moving objects disperse from common locations or areas (Dodge et al. 2008; Fang et al. 2017). This pattern is often observed after convergence or aggregation. Similar to convergence, divergence movement patterns can also be distinguished by direction constraints, as shown in Fig. 14. Formally, a divergence pattern is defined as a set of moving objects $\boldsymbol{DP} \in \boldsymbol{O}$ that satisfy the following conditions:

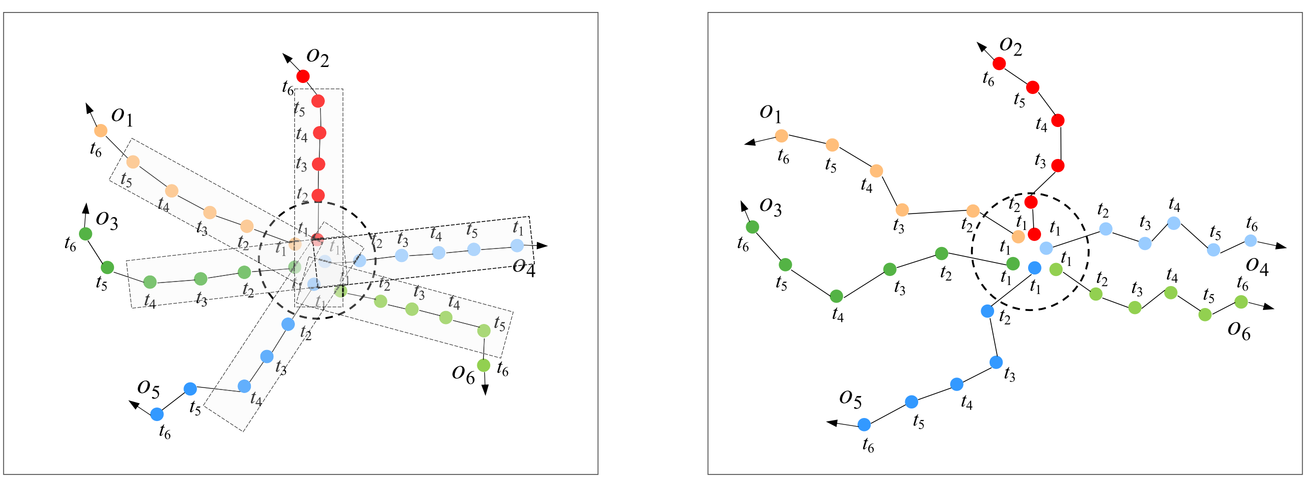
- Direction constraint: The movement direction of any moving object in ***DP*** can either remain largely unchanged over time or the moving objects in ***DP*** are scattered in different directions, and some moving objects are allowed to share similar directions.
- Common origin: Moving objects in ***DP*** originate from a common region ***R*** at the initial time.
- Direction diversity: The moving objects in ***DP*** should be dispersed from the common region in diverse directions, while some objects are allowed to share similar movement directions.
- Asynchronous dispersion: Moving objects in ***DP*** are allowed to leave the region R at different times.

When moving objects scatter from the common location simultaneously, this divergence pattern is called a *breakup* (Dodge et al. 2008), which corresponds to the opposite pattern of the *encounter*.

Co-location Movement Patterns

The co-location movement pattern is a spatiotemporal phenomenon where two or more moving objects frequently maintain spatial proximity (Andrienko and Andrienko 2007; Evans et al. 2016; Maiti and Subramanyam 2021). Accompanying movement patterns can be considered a specialized type of co-location, which incorporates additional constraints (e.g., directional consistency, movement synchronization, and group membership). While all accompanying patterns are co-location patterns, the reverse is not necessarily true, as co-location may not be persistent across all locations.

Co-location movement patterns can be categorized into synchronized and lagged co-location patterns. As shown in Fig. 15, synchronized co-location describes instances where moving objects remain spatially proximate without significant temporal offset. In contrast, a lagged co-location occurs



(a) (b)

Fig. 14 Illustration of the divergence movement pattern. **a** Moving objects o_1, o_2, o_3, o_4, o_5, and o_6 maintain largely constant directions during their divergence process, and **b** moving objects o_1, o_2, o_3, o_4, o_5, and o_6 disperse to different directions, showing noticeable directional deviations during their movements

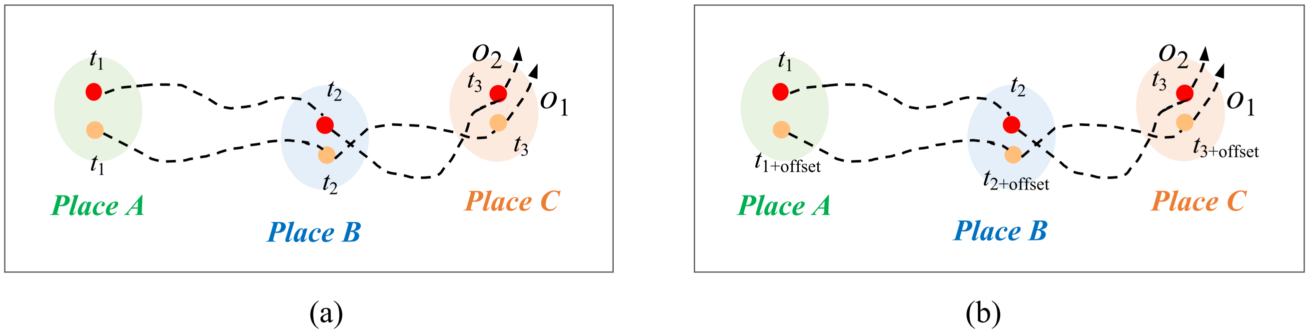


Fig. 15 Illustration of the co-location pattern. **a** Synchronized co-location, where moving objects o_1 and o_2 are present at places *A*, *B*, and *C* at the same time, and **b** lagged co-location, where moving objects o_1 and o_2 visit the same location at different times

when moving objects maintain spatial proximity, but with a temporal lag or decay.

Frequent Movement Patterns

Frequent movement patterns capture statistically significant repetitions in movement behaviors exhibited by one or multiple moving objects, such as recurring sequences, paths, or regions (Liu et al. 2011; Da Silva et al. 2014; Sweeting et al. 2017; Yu 2018). While frequent movement patterns can be considered a specific type of co-location pattern when the frequency of co-location exceeds a predefined frequency threshold, they are broader in scope. Frequent patterns encompass both individual and collective movements. In contrast, co-location movement patterns primarily reveal collective movements. Formally, given a dataset $\mathscr{T}=\{T_1, \ldots, T_n\}$ from objects $\boldsymbol{O}$, $\boldsymbol{F}$ (σ, ϵ) is a frequent movement pattern that satisfies the following conditions:

- Support thresholdconstraint: The frequency of occurrence of $\boldsymbol{F}$ among the movements (e.g., sequence, path, and region) must exceed a predefined support threshold σ, where $0 \leq \sigma \leq 1$, i.e., $|\{T_i \in \mathscr{T} | \boldsymbol{F} \preceq T_i\}|/|\mathscr{T}| \geq \sigma$, where $\preceq$ denotes pattern occurrence (exact or approximate).
- Similarityconstraint: Any movement instance within $\boldsymbol{F}$ must exhibit a high degree of similarity, i.e., $\forall$ F_k, F_l instances of $\boldsymbol{F}$, Sim $(F_k, F_l) \geq \epsilon$, where Sim(.) is a similarity metric (e.g., DTW, Hausdorff distance) and ϵ represents the predefined similarity threshold.

Frequent sequences and paths are two common frequent movement patterns. The former focuses on recurring sequences of movement behaviors such as location visits (A $\rightarrow$ B $\rightarrow$ C), regardless of the exact spatial paths between locations. At the same time, the latter emphasizes repeated spatial trajectories or paths with geometric similarity, such as commonly traveled routes or ecological corridors.

Periodic Movement Patterns

Periodic movement patterns describe temporally regular repetitions of movements exhibited by moving objects (Li et al. 2010b; Li et al. 2012). Like frequent movement patterns, they can reflect individual and collective movements. However, frequent patterns emphasize spatial recurrence, while periodic patterns emphasize temporal regularity (e.g., daily commuting and annual migrations). Formally, given a dataset $\mathscr{T}=\{T_1, \ldots, T_n\}$ (e.g., path, location visits, trajectory segments) from objects $\boldsymbol{O}$, $\boldsymbol{P}$ (τ, ϵ, σ) is a periodic pattern that satisfies the following conditions:

- Temporal periodicityconstraint: The pattern $\boldsymbol{P}$ recurs at regular time intervals τ, where τ represents the period length, i.e., $\forall$ instance $P_i \in \boldsymbol{P}$ occurring at time t_i, $\exists$ instance P_j occurring at time t_j such that $|t_j - t_i| = k \cdot \tau$, where $k \in \mathbb{Z}^+$.
- Similarity constraint: Each instance of $\boldsymbol{P}$ must be spatially and temporally similar to other instances of $\boldsymbol{P}$, i.e., $\forall$ P_i, P_j instances of $\boldsymbol{P}$ where $|t_j - t_i| = k \cdot \tau$, $\text{Sim}(P_i, P_j) \geq \epsilon$, where Sim(.) is a similarity metric and ϵ represents the predefined similarity threshold.
- Minimum occurrence constraint: The pattern $\boldsymbol{P}$ must occur at least σ times within the observation period, where σ is the predefined minimum repetition threshold.

The periodicity in periodic movement patterns can be fixed, variable, multi-scale, partial, and even hierarchical. Figure 16 illustrates the periodic movement pattern with a single fixed period.

Abnormal Movement Patterns

Abnormal movement patterns describe the movements of moving objects deviating from their expected or normal behaviors (Wu et al. 2021; Shi et al. 2022a). These deviations can manifest as abnormal changes in movement

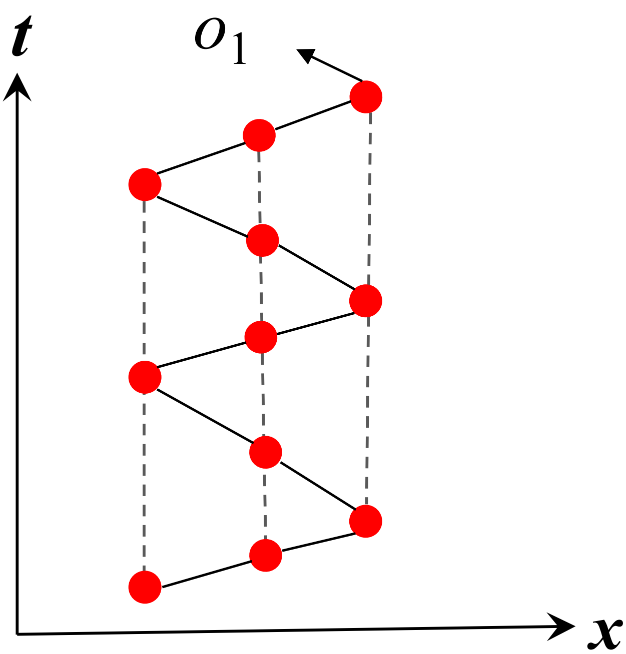


Fig. 16 Illustration of a single-period movement pattern

parameters (e.g., speed and direction), irregularities in spatial positions (e.g., unexpected stops, detours, or deviation from established routes), or anomalies in temporal dynamics (e.g., unexpected delays or disappearance, and irregular timing), as presented in Fig. 17. Formally, given a dataset $\mathcal{T} = \{T_1, \ldots, T_n\}$ from a set of objects $\boldsymbol{O}$, $\boldsymbol{A}$ (θ) is an abnormal movement pattern that satisfies the following conditions:

Deviation constraint: The pattern $\boldsymbol{A}$ exhibits a quantifiable deviation from an established baseline $\boldsymbol{B}$ that exceeds a predefined deviation significance threshold θ, i.e., Deviation $(\boldsymbol{A}, \boldsymbol{B}) \geq \theta$, where Deviation(.) quantifies the deviation (e.g., dissimilarity) of $\boldsymbol{A}$ from $\boldsymbol{B}$.

Methodologies for Movement Pattern Mining

Movement pattern mining aims to recognize valid, novel, practical, and ultimately understandable non-random properties and relationships in movement data (Laube 2014). To date, a wide variety of algorithms have been developed. According to existing research, methodologies for movement pattern mining can be broadly categorized into the following ten approaches:

Rule-Based Pattern Mining

Rule-based pattern mining methods employ explicitly predefined logical rules or constraints to identify movement patterns (Michalski 1980; Hassan et al. 2010). This approach defines a pattern as a set of conditions that moving objects must satisfy within specific spatial regions or time intervals. Rule $R = \{r_1, r_2, \ldots, r_m\}$ encodes domain-specific pattern knowledge, often represented as logical predicates that evaluate spatial relations, temporal sequences, and motion properties. Formally, given a set of moving objects $O = \{o_1, o_2, \ldots, o_n\}$, a rule in R can be expressed as follows:

$r_i(o, S, T, A)$: $C_1 \wedge C_2 \wedge \cdots \wedge C_k \rightarrow P$

where o represents the moving object, S represents the spatial domain, T represents the temporal domain, A is the set of motion attributes, C_j denotes a condition, and P is the pattern to be recognized. For example, a rule might state: if (velocity > 5 m/s) and direction changes > 30° in 20 s, and (acceleration > 4 m/s^2), then classify the movement as anomalous.

Rule-based pattern detection can be regarded as an inductive reasoning problem guided by specific rules (Michalski 1980). Common rule reasoning approaches include forward and backward chaining (Al-Ajlan 2015), the Rete algorithm

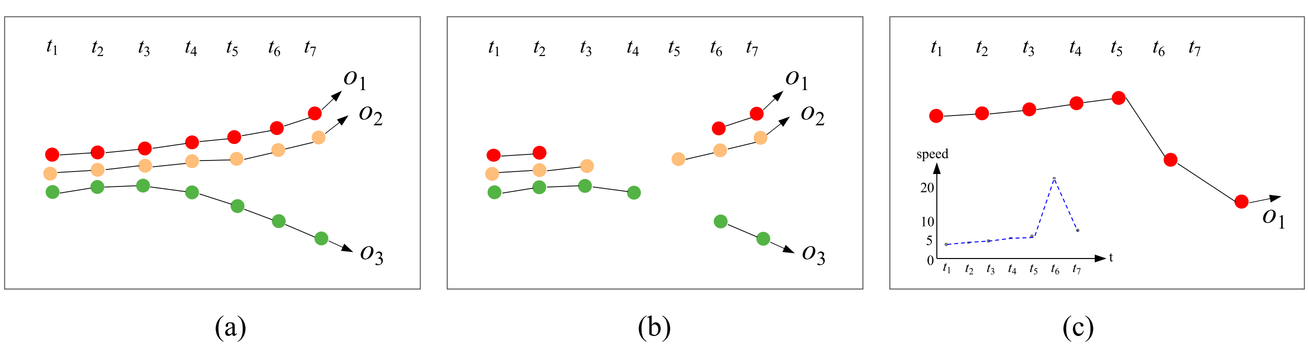


Fig. 17 Illustration of abnormal movement patterns. **a** Spatial path deviation, where moving object o_3 diverges from the shared trajectory of o_1 and o_2 starting at t_3; **b** temporal interrupt, where o_1 is inactive from t_2t5t_2 to t_5, o_2 from t_3 to t_5, and o_3 from t_4 to t_6; and **c** speed surge and slowdown of o_1

(Forgy 1989), decision trees (e.g., ID3, C4.5, and CART) (Arentze et al. 2000), pattern trees (Huang et al. 2008), fuzzy pattern trees (Senge and Hüllermeier 2010; Shaker et al. 2013), and spatial rule-based engines (Kim et al. 2018). Rule-based approaches are practical when movement patterns adhere to well-established norms and can be explicitly modeled, offering high interpretability with each detection traceable to specific rules. However, these methods struggle with complex or novel movement patterns that are difficult to express as explicit rules.

Notably, with the advent of large language models (LLMs) (Chang et al. 2024), semi- or fully automated rule generation from existing extensive datasets and knowledge bases has become increasingly feasible (Zhu et al. 2023). This paradigm shift revolutionizes traditional, expert-driven rule formulation for movement pattern detection (Zhang et al. 2024b; Zhou et al. 2024).

Template Matching-Based Pattern Mining

Template matching is one of the most classical pattern recognition methods (Jain et al. 2000), which identifies movement patterns by comparing observed movement sequences with predefined templates (or prototypical patterns). Templates can take various forms, such as geometric templates (Frank et al. 2012; Zhang et al. 2021), state transition matrix (Chaudhry et al. 2013), or probability graphs (Yuan et al. 2016). Templates can be manually designed by domain experts or automatically extracted using machine learning techniques such as clustering. For example, maritime routes extracted from historical ship trajectories can serve as reference templates for detecting abnormal vessel behaviors (Soleimani et al. 2015; Shi et al. 2022a). Another typical example is the Relative Motion (REMO) research in GIScience. The templates are predefined to match and identify specific configurations and arrangements in the REMO matrix constructed from motion parameters (e.g., speed, direction, and acceleration) (Laube and Imfeld 2002; Laube et al. 2005; Fisher et al. 2005). Unlike rule-based methods that impose strict logical constraints, this approach is more flexible by quantitatively measuring the similarity between the observed movement and the template (Fu et al. 2007). Various similarity measures, such as dynamic time warping (DTW) (Berndt and Clifford 1994; Zhao and Itti 2018), Fréchet distance (Eiter and Mannila 1994), and graph edit distance (Gao et al. 2010), can be used to quantify how closely an observed movement matches the given template.

Templating match-based pattern mining methods are commonly used in military surveillance (e.g., identifying irregular patrols and tactical maneuvers), gesture recognition, sports analysis, and maritime monitoring (e.g., detecting suspicious ship movements). However, this method requires significant computing resources for similarity calculation. Its effectiveness also depends on the quality and completeness of the template library, which can be challenging for rare or unseen patterns.

Clustering-Based Pattern Mining

Clustering-based pattern mining employs clustering techniques to detect inherent structures or cohesive moving object groups from movement data (Li et al. 2004; Kalnis et al. 2005; Yuan et al. 2017a; Peng et al. 2025). This approach distinguishes between *static* movement patterns that only consider spatial proximity and *dynamic* movement patterns that consider both spatial and temporal information.

Static movement pattern mining is categorized into *point-based*, *trajectory segment-based*, and *trajectory-based* methods based on the clustering unit. The difference among these three methods primarily lies in the similarity measures they adopt. Point-based methods generally use Euclidean or network distances to measure the spatial proximity between points (Madhulatha 2012). In contrast, trajectory-based and trajectory segment-based methods access the similarities between complete trajectories or segments using measures such as DTW, DSPD (Peng et al. 2024), Hausdorff, and Fréchet distance. Trajectory segment-based clustering, such as the partition and group framework (Lee et al. 2007), must divide trajectories into segments before clustering. It enables the discovery of local movement patterns that global trajectory clustering may overlook.

Dynamic movement pattern mining generally adopts two strategies: temporal slicing with cluster tracking or spatiotemporal unified clustering. The former implements clustering independently at each timestamp to identify spatially proximate groups of moving objects and then connects these clusters across consecutive timestamps to discover moving clusters (Kalnis et al. 2005). The latter directly incorporates the temporal dimension into the clustering process. By measuring the spatial and temporal similarity between entire trajectories or segments, it can detect movement patterns over space and time (Nanni and Pedreschi 2006; Ansari et al. 2020).

Regarding the specific clustering algorithms used for partitioning movement data, they can be broadly classified into six types: density-based, hierarchical-based, partition-based, model-based, graph-based, and grid-based methods (Saxena et al. 2017). Among them, density-based clustering algorithms (e.g., DBSCAN, HDBSCAN) are the most widely used because they can discover clusters of arbitrary shapes without requiring a predefined number of clusters (Bhattacharjee and Mitra 2021). However, the choice of similarity measures and clustering parameters may significantly influence their effectiveness.

Sequential Pattern Mining

Sequential pattern mining (SPM) extracts temporally ordered patterns from movement data by identifying frequently occurring subsequences, emphasizing the temporal order and frequency of movements (Mazumdar and Sarma 2024). SPM was first formally proposed by Agrawal and Srikant (1995) to discover frequent sequences of item sets in transaction databases, which extends their earlier research on frequent pattern mining (FPM) (Agrawal and Srikant 1994). While traditional FPM only identifies co-occurring items (as in market basket analysis) without considering the order of items, SPM explicitly incorporates temporal or order constraints. SPM has been widely used in movement analysis, such as identifying migration routes (Shaw and Gopalan 2014) and human mobility routines (Sweeting et al. 2017) (a series of visited locations over time, like "Home → Office → Gym").

Several studies have systematically reviewed SPM algorithms (Mabroukeh and Ezeife 2010; Fournier-Viger et al. 2017; Gan et al. 2019; Mazumdar and Sarma 2024). Based on existing research, we categorize SPM algorithms into Apriori-based and pattern growth-based approaches. Apriori-based methods, such as AprioriAll and AprioriSome (Agrawal and Srikant 1995), GSP (Srikant and Agrawal 1996), SPADE (Zaki 2001), and SPAM (Ayres et al. 2002), extend the original Apriori algorithm (Agrawal and Srikant 1994) to sequential data by iteratively generating and pruning candidate sequences based on the Apriori property that all subsequences of a frequent sequence must also be frequent. In contrast, pattern growth-based methods, including FP-growth (Han and Pei 2000), Free Span (Han et al. 2000b), Prefix Span (Han et al. 2001), FP-tree (Han et al. 2004), and their extensions (Saraf et al. 2015; Borah and Nath 2018; Pan et al. 2018; Niyazmand and Izadi 2019), employ the divide and conquer strategy to directly grow frequent patterns recursively without generating candidate sequences.

While SPM can discover meaningful sequential movement patterns, its effectiveness is related to sequence representation methods (i.e., how to transform raw movement data into sequences), parameter settings (e.g., minimum support thresholds), and noise handling. Additionally, the computational complexity of SPM algorithms is also challenging for large-scale movement data, as the search space expands exponentially with sequence length. Recent research has focused on parallel and distributed SPM algorithms to address the scalability issues of SPM algorithms (Gan et al. 2019).

Association Rule-Based Pattern Mining

Association rule mining (ARM) is another key extension of frequent pattern mining, which complements SPM by focusing on implication relationships or rules rather than temporal sequences (Kumbhare and Chobe 2014; Yu 2018). ARM can discover probabilistic rules between different spatial locations or events. For example, if tourists visit Place *A*, then they are also likely to visit Place *B*, or if a taxi picks up passengers downtown in the morning, then it is expected to get off near business districts.

Support, confidence, and lift are commonly used metrics in ARM and SPM to evaluate the significance of discovered associate rules or frequent subsequences. The main difference is that SPM explicitly incorporates temporal or ordering constraints, while ARM focuses on mining directional relations between items or events regardless of their order. ARM also faces challenges similar to those of SPM and requires careful preprocessing, parameter tuning, and efficiency optimization (Soni et al. 2016; Fister et al. 2023).

Periodic Pattern Mining

Periodic pattern mining (PPM) identifies repetitive movement behaviors at regular intervals (Cao et al. 2007; Li et al. 2010b; Li et al. 2012). Cyclic phenomena are in human and natural systems, such as daily commuting routes, seasonal migration paths, and tidal movements. The key challenge in PPM is the determination of appropriate periodicity (Edelhoff et al. 2016). The presence of multiple temporal scales (Yuan et al. et al. 2017b), locality (Fournier-Viger et al. 2021), noise, and irregular sampling (Trajcevski et al. 2004; Wang et al. 2015) in movement behaviors complicates the identification of periodic movement patterns. Current PPM approaches can be divided into four categories:

- *Frequency-based methods* transform movement data into the frequency domain to detect periodic components by using the Fourier transform, wavelet analysis, or their derivatives (Li et al. 2010b; Sirisha et al. 2013). This type of method is robust to noise. It can effectively identify dominant periodicities across long-term spans but is sensitive to sampling irregularities and limited in detecting irregular periodicity.
- *Temporal-based methods* use temporal correlations to detect intervals and synchronization behaviors (Sirisha et al. 2013). For example, autocorrelation analysis measures self-similarity across different time intervals (Breitenbach et al. 2023). This approach is easy to understand and implement but is sensitive to noise and irregular data and struggles with complex periodic patterns.
- *Sequence-based methods* identify repeated subsequences based on SPM techniques. To enhance efficiency, various data structures such as suffix trees (Nishi et al. 2013; Rasheed and Alhajj 2010) and suffix tris (Chanda et al. 2015), and list-based structures (Kim et al. 2020) are

commonly used. Motifs, defined as repeated segments, serve as basic units for periodic pattern discovery (Mueen 2014; Su et al. 2020). This method is suitable for detecting local and partial repetitions but is often computationally intensive and requires discretizing trajectories into symbolic sequences.

- *Model-based methods* employ statistical or clustering models to identify periodic patterns. These methods are robust to noise and effective for complex periodicity. For example, Li et al. (2016) combined kernel density estimation with the chi-square test to detect mixed periodic patterns. Zhang et al. (2018) extended the Traclus algorithm to hierarchical clustering to reveal hierarchical periodicity. However, they rely heavily on the initial model assumptions, which may affect reliability.

In practice, these approaches can be combined to enhance the performance of periodicity detection. For example, Li et al. (2010b) used the Fourier transform and autocorrelation to detect multiple periods and then applied a probabilistic model to capture specific periodic patterns.

Anomaly Detection-Based Pattern Mining

Anomaly detection identifies abnormal movement behaviors that deviate from expected patterns based on the assumption that abnormal patterns exhibit measurable differences in spatial, temporal, spatiotemporal, or motion characteristics compared to regular patterns. It provides valuable insights into abnormal events, potential threats, or unusual activities. Deng et al. (2016) classified anomaly detection methods as statistics-based, distance-based, density-based, angle-based, and clustering-based approaches. Shi et al. (2022a) divided anomalies into spatial, spatiotemporal, and spatial-semantic attribute anomalies. Based on existing studies, we classify abnormal movement pattern detection methods into three primary categories.

- *Spatiotemporal trajectory analysis-based detection* analyzes the characteristics of movement data to identify anomalies (Cao et al. 2017; He et al. 2019). Visualization and statistical analysis are two commonly adopted techniques. The former can intuitively reveal the anomalies in data, such as path deviation, disappearance, and accidental gathering. The latter excels in finding irregularities of movement attributes, such as abrupt changes, but is constrained by predefined thresholds.
- *Rule-based anomaly detection* employs predefined rules or constraints to detect anomalies. For example, in urban traffic monitoring, rules can be speed and turn limits on specific roads. If the vehicle's behavior violates these rules (e.g., speeding or taking an illegal turn), it will be marked as an anomaly. Detailed rule-based pattern detection is described in the "Rule-Based Pattern Mining" section.
- *Clustering-based anomaly detection* considers anomalies as dense clusters that represent unusual gatherings, or as outliers that do not belong to any cluster. For example, Farahnakian et al. (2023) employed DBSCAN to detect illegal ship activities. Qin et al. (2021) used clustering to detect abnormal weather patterns from large-scale hurricane trajectory data. Detailed clustering-based pattern detection methods can be found in the "Clustering-Based Pattern Mining" section.

This method often complements other movement pattern mining techniques. For example, sequential pattern mining can establish baseline normal behaviors against which anomalies are detected, or directly identify anomalous sequences of movements.

Graph-Based Pattern Mining

Graph-based pattern mining uses graph structures to represent movement data, which differs from previously discussed methods that rely on discrete spatial point-based, vector-based, or sequence-based representations. This approach explicitly transforms movement data (e.g., trajectories) of moving objects and their relationships (e.g., spatial proximity, temporal correlation, attribute similarity, or semantic connections) into nodes and edges within a graph structure. Then, algorithms from graph theory can be applied to detect certain movement patterns that conform to specific topological or relational constraints. For instance, Tritsarolis et al. (2021) introduced a graph-based co-movement pattern framework, called EvolvingClusters, which utilizes Maximal Cliques and Maximal Connected Subgraphs to detect flocks and convoys collectively. Similarly, Zhang and Van de Weghe (2024) employed Reeb graphs constructed from movement data to identify moving flock patterns.

Graph-based pattern mining can model the evolution of movement patterns such as emergence, expansion, persistence, splitting, and dissolution. It draws upon established graph-based community detection methods that have proven effective in tracking the evolution of social, biological, or other networks (Cordeiro et al. 2016; Li et al. 2023b). Recent advances also demonstrate its potential to predict future movement patterns (Tritsarolis et al. 2024). However, its effectiveness depends on mapping movement pattern concepts (e.g., flocks, convoys) onto graph-theoretic constructs (e.g., cliques), which may limit its applicability in discovering complex and diverse movement patterns.

Query-Based Pattern Mining

Query-based pattern mining aims to extract movement patterns by customizing specific queries (Kalnis et al. 2005; Yadamjav et al. 2020). It enables targeted identification of movement patterns by filtering the search range instead of generating all patterns. For example, a user might query for "all pedestrian groups of three or more who walk for 10 min within a 20-m radius of the city center between 11–12 am." Common query types for movement pattern detection include spatiotemporal range queries (Helmi and Banaei-Kashani 2017), KNN queries (Bareche and Xia 2022), entity-based queries (Cheng et al. 2009), trajectory similarity queries (Chen et al. 2005), and semantic attribute queries (Bogorny et al. 2009). They can be combined for more complex pattern retrieval.

The effectiveness of query-based pattern mining largely depends on efficient data structures and indexing mechanisms that support fast retrieval of movement patterns. Standard indexing techniques include R-tree (Yadamjav et al. 2020), KD-trees (Orakzai et al. 2021), grid-based indexes (Vieira et al. 2009), inverted indexes (Tanaka et al. 2016), and hybrid indexes (Chen et al. 2019; Bareche and Xia 2022), which have been used in several studies to enhance the efficiency of pattern discovery (Kalnis et al. 2005; Yadamjav et al. 2020). Query-based pattern mining is more efficient and avoids mining all possible movement patterns on the whole dataset. However, designing effective queries requires users to understand movement patterns and their characteristics deeply.

Deep Learning-Based Pattern Mining

Deep learning-based pattern mining employs advanced neural network architectures to extract movement patterns (Wang et al. 2020; Yang and Gidófalvi 2020; Kleanthous et al. 2022). This approach represents a paradigm shift in movement pattern mining from explicit rule-based to implicit data-driven models. Through end-to-end training, this approach learns latent movement feature representations and implicit pattern mappings from raw movement data. It enables the recognition of more complex movement patterns that are difficult to describe explicitly and can achieve the collaborative detection of multiple coexisting movement patterns. For example, Yang and Gidófalvi (2020) converted trajectories into images and used a convolutional neural network to recognize and classify different movement patterns. Yu and Huang (2022) developed an end-to-end deep learning framework (STDTB-AD) to detect abnormal taxi trajectory patterns and driving behaviors. Zhou et al. (2025) achieved efficient spatial co-location pattern mining by leveraging advanced representation learning techniques to learn and integrate embedded representations of local context information.

Although deep learning-based movement pattern mining is still at a relatively early stage, the powerful representation capabilities of deep neural networks and their ability to model complex nonlinear relationships underscore their potential in movement pattern mining. However, while this approach exhibits great potential, it also brings technical challenges. These challenges are also the focus to be overcome in the future to promote the application of deep learning in the movement pattern mining field.

- Data requirements: Deep learning models typically require large amounts of labeled movement data for effective training and learning. However, such datasets are often scarce and expensive to obtain, as they demand substantial human effort, material resources, and financial investment. This remains one of the primary obstacles to the widespread application of deep learning in movement pattern mining.
- Feature learning: Movement patterns are diverse and complex. Effective feature learning should capture the individual movement dynamics and inter-object relationships that arise from spatial, temporal, and relational dimensions. This requires the design of specialized network architectures that can model such interactions and dependencies among moving objects.
- Computational cost: Training and deploying deep learning models, especially recurrent architectures (e.g., LSTM and GRU) and attention-based models, or graph-based models, can be computationally expensive in terms of time and resources. This may limit the applicability of deep learning-based pattern mining in real-time or resource-constrained environments.
- Interpretability: The inherent black-box nature of neural networks makes it difficult to interpret or validate the discovered movement patterns. In contrast, traditional approaches introduced above, such as clustering and template matching, offer greater transparency, enabling explicit reasoning behind pattern detection.

Applications of Movement Pattern Analysis

Moving objects are constantly in motion, generating vast amounts of movement data. The mining and analysis of movement patterns are beneficial in various application domains. A few of the extensive applications are outlined below.

Transportation and Urban Planning

In urban environments, vehicles and pedestrians continuously interact with transportation networks, generating diverse movement patterns that can reflect city dynamics, traffic efficiency, and infrastructure utilization. Analyzing these patterns can provide crucial information for optimizing traffic management and urban planning. For example, real-time monitoring of these movement patterns enables authorities to understand traffic conditions, detect emerging traffic issues (e.g., traffic congestions, accidents, and abnormal traffic flows), and implement timely countermeasures (Raiyn and Toledo 2014; Khazukov et al. 2020; Shanthappa et al. 2023). By integrating discovered movement patterns with additional information (e.g., road networks, land use, and POIs), it can further identify root causes of transportation problems, predict future traffic demands, and optimize infrastructure utilization and service allocation (Yue et al. 2021; Liang et al. 2023).

Human Movement and Behavior Analysis

As the main society entities, humans exhibit complex and diverse movement patterns while engaging in various social, economic, and cultural activities (Gonzalez et al. 2008; Parent et al. 2013; Vandecasteele et al. 2014). Mining and analyzing these movement patterns can provide valuable insights into individual and collective behaviors, spatial interactions and preferences, activity choices, and potential group events. For example, identifying accompanying patterns from individual activity trajectories can discover groups moving together in geographic space (Guo et al. 2014), which can serve applications such as group division and carpooling. Recognizing gathering patterns can reveal potential group events (Zheng et al. 2013) and inform public safety and incident response.

Ecology and Animal Behavior

Animals and their movements have long been the focus of research in biology, ecology, and conservation sciences. With the development of positioning technology and big data analysis technology, animal movement research has changed from an early ecological movement to an emerging movement ecology paradigm (Nathan et al. 2008, 2022; Joo et al. 2022). There are a variety of movement patterns in animals, which can reflect various aspects of animal behavioral adaptation, physiological needs, ecological interaction, and response to the environment. For instance, detecting daily activity ranges and paths of animals can reflect hunting, foraging, drinking, and other habitual behaviors (Bennison et al. 2018). The mining and analysis of movement patterns can help understand animal behaviors and support related practical applications.

Maritime Management and Safety

Modern maritime transportation systems continuously monitor and report vessel movements across global waterways through the Automatic Identification System (AIS), satellite-based monitoring, and coastal radar networks. The massive ship movement data contains valuable spatiotemporal information about maritime traffic patterns and ship behaviors. Mining patterns from these data is becoming increasingly crucial in maritime domains to improve maritime safety and operational efficiency, such as route optimization, collision risk assessment, and illegal activity detection (Soleimani et al. 2015; Wen and Yan 2019; Kontopoulos et al. 2021; Chen et al. 2023). For example, maritime authorities can establish standardized shipping lanes based on historical ship trajectories and predict congestion in busy waterways (Stergiopoulos et al. 2018; Tu et al. 2020).

Military Surveillance

The real-time tracking and movement pattern analysis of military targets helps understand enemy behaviors and intentions (Fileto et al. 2015), improving battlefield situational awareness and assisting military decision-making. For example, frequent maritime movement patterns may reveal critical defensive areas, base locations, and strategic chokepoints, while aggregation movement patterns observed in land military targets may indicate encircling maneuvers or concentration of force.

Environment Monitoring and Climate Analysis

Movement pattern mining and analysis can also be extended to certain geographical phenomena or dynamic environmental processes. In atmospheric and oceanic sciences, studying the movement patterns of key variables such as air masses, ocean currents, or weather systems is essential. For example, mining cluster patterns from hurricane trajectories can reveal recurring storm paths and high-incidence areas (Lim et al. 2016). Mining these movement patterns and correlating them with environmental conditions can provide insights into the underlying causes of natural phenomena and improve predictive accuracy and early warning effectiveness (Tsai et al. 2009; Dong and Pi 2013).

Sport Science

The above applications, whether involving visible or abstract moving objects, are grounded in geographic coordinate systems, within which spatiotemporal activities can be mapped, and analysis happens in open-world environments. The movement pattern analysis of human kinematic activities that occur within confined areas is also valuable for applications in sports sciences. The trajectories obtained through motion capture systems, wearable sensors, and optical tracking technologies make it possible to analyze athlete movements at individual and team levels. Mining movement patterns from sports movement data can help identify abnormal biomechanical patterns, recognize fatigue indicators, and discover tactical patterns, thus optimizing athlete performance (Sweeting et al. 2017; Van der Kruk and Reijne 2018).

Others

Movement pattern mining and analysis also have critical applications in logistics management (Zhao et al. 2020b), epidemic mobility (Shi et al. 2022b), security surveillance (de Lucca Siqueira and Bogorny 2011), tourism planning (Zhong et al. 2019), and eye movement analysis (Hindmarsh et al. 2021).

Open Movement Datasets and Tools for Movement Pattern Mining

Open resources are essential to promote reproducible research on movement pattern mining. This section introduces available and widely used movement datasets and tools or toolkits/software relevant to this field.

Movement Datasets

Available movement datasets are essential for developing various spatiotemporal pattern mining algorithms (Zheng 2015). The development of mobile sensing and positioning technologies has provided unprecedented opportunities for tracking and collecting the movement of diverse moving objects such as humans, vehicles, animals, ships, and even natural phenomena such as hurricanes. These movements are usually recorded as sequences of trajectory points, which capture rich behavioral regularities and movement patterns of moving objects in various domains. In addition to real datasets, simulated datasets play an essential role, especially when real data is scarce, incomplete, or unavailable due to privacy or other constraints (Kapp et al. 2023). Table 1 presents several representative real-world movement datasets and movement simulators along with their brief introductions.

Tools for Movement Data Exploration and Movement Pattern Mining

Many open-source libraries, platforms, or software tools have been developed to support spatiotemporal data processing, exploratory analysis, pattern mining, and visualization. These tools significantly lower the barrier to entry for researchers and practitioners. One of the earliest systems in this field is *MoveMine*, which supports the detection and analysis of various movement patterns such as periodicity, flocks, following, and attraction/avoidance relationships (Li et al. 2011; Wu et al. 2014). Later, Graser (2019) developed the first dedicated Python library for movement data exploration and analysis, providing a range of functions such as stop detection, trajectory generalization, trajectory aggregation, trajectory cleaning, and smoothing. Building on these foundations, some Python and R packages and database extensions have been developed to support more comprehensive and flexible movement data analysis. Notable tools include *Scikit-Mobility* (Pappalardo et al. 2022), *Stmove* (Seidel et al. 2019), *Traja* (Shenk et al. 2021), *Trackintel* (Martin et al. 2023), *PyMove*, and *MobilityDB* (Zimány et al. 2020).

In addition, several desktop GIS software tools are also widely adopted for visualizing, managing, and analyzing movement data. These tools can also help prepare movement datasets for further mining in specialized settings.

Challenges and Future Directions

Despite numerous studies on movement pattern mining and analysis, several challenges remain and need to be addressed, which also represent potential future research directions.

Spatial and Temporal Scale Issues

Movement patterns are scale-dependent, with different patterns possibly emerging at varying spatial and temporal scales (Kays et al. 2023). Identifying multi-scale movement patterns is challenging, and it requires determining appropriate spatial and temporal scales manually or by adaptive algorithms and designing effective algorithms to capture movement patterns across these scales. Selecting an optimal spatiotemporal resolution is critical, as certain patterns may only be discernible at specific scales. It is necessary to balance the trade-off between granularity loss at coarse scales and noise amplification at finer scales when utilizing hierarchical modeling approaches.

Table 1 Representative real-world movement datasets and movement simulators

Dataset	Description	Object type	Scale (objects/trajectories)	Source	References
Geolife	GPS trajectories of human daily activities collected in Beijing, China (2007–2012)	Humans	182 users, 17,621 trajectories	Microsoft Research	Zheng et al. (2010)
T-Drive	Large-scale GPS trajectories of taxis collected in Beijing, China (Feb–Apr 2008)	Taxis	33,000 taxis, 4.96 million trajectories	Microsoft Research	Yuan et al. (2010)
Movebank	Open repository for global animal tracking data from GPS collars and sensors	Animals (e.g., birds, fish, mammals)	Varies by study (1000 + species)	Max Planck Institute	-
Marine Cadastre	Vessel AIS tracking data in U.S. waters (2009–present)	Ships	Millions of vessel tracks	NOAA, BOEM, NAVCEN	-
IBTrACS	Historical hurricane paths over the past 150 years (global)	Hurriances	6000 + cyclones	NOAA	Knapp et al. (2010); Gahtan et al. (2024)
HURDAT2	Best hurricane track data (1851–present)	Hurriances	1500 + cyclones	NOAA	Landsea and Franklin (2013)
OpenstreetMap trajectories	Crowdsourced GPS traces (global)	Humans/vehicles	Millions of trajectories	OSM	-
Outdoor activity tracks	Crowdsourced outdoor activity trajectories	Humans	Varies by data platform	2bulu/ Outdooractive	-
Porto taxi dataset	One year of taxi trips with GPS data in Porto, Portugal (2013–2014)	Taxis	442 taxis, 1,710,671 trajectories	Kaggle/UCI	Moreira-Matias et al. (2013)
Simultaed trajectory dataset	Standard benchmark dataset for trajectory clustering simulating urban road traffic scenarios	Vehicles	Cross: 1900, I5sim: 800, I5sim2/I5Ssim3: 1600, I5: 806, Labomni: 209	-	Morris and Trivedi (2009)
Brinkhoff Generator	Generator for simulating object movements on networks	Simulated objects	User-defined	-	Brinkhoff (2002)
BerlinMOD	Benchmark for moving object data simulation and analysis in Berlin, Germany	Simulated objects	User-defined	-	Düntgen et al. (2009)
SUMO	Open-source traffic simulator for urban mobility scenarios	Simulated objects	User-defined	Eclipse Foundation	Behrisch et al. (2011)
Ditras	A trajectory generator to simulate spatiotemporal patterns of human mobility	Simulated objects	User-defined	-	Pappalardo and Simini (2018)
TrajGDM	A trajectory generation framework for simulating human mobility	Simulated objects	User-defined	-	Chu et al. (2024)

Heterogeneous Data Fusion

Movement data of moving objects may come from various sources (e.g., GPS, RIF, IoT sensors) with varying spatiotemporal resolutions, formats (e.g., trajectory, image), and coordinate systems (WGS84 vs. local grids). It is challenging to accurately identify movement patterns from such multi-source, heterogeneous movement data. Future research should prioritize the development of robust data fusion frameworks to generate more comprehensive and high-quality movement data, which is essential for the reliable and accurate extraction of movement patterns.

Coupled Movement Pattern Mining

Moving objects may exhibit multiple movement patterns during their movements, which poses great challenges in two aspects: (1) how to identify different movement patterns across various temporal ranges (i.e., the evolution of movement patterns), and (2) how to determine the dominant patterns when multiple movement patterns coexist. Current research typically focuses on mining specific movement patterns, failing to fully characterize the movement regularities of moving objects. Future research should develop approaches that can identify multiple and primary movement patterns.

Online Movement Pattern Mining

The continuous accumulation of streaming movement data has created urgent demands for efficient and online movement pattern mining. Existing movement pattern mining algorithms often have high time complexity, making real-time pattern detection challenging. To balance algorithm efficiency and detection accuracy in dynamic environments, it is necessary to develop efficient and effective algorithms for online detection of movement patterns and their changes in future work.

Standard Movement Pattern Dataset and Code Benchmarks

The availability of standard benchmarks for movement patterns is essential to ensure the reproducibility and comparability of research, such as evaluating the performance of diverse pattern mining algorithms. Although a variety of publicly available movement datasets exist (as introduced in the "Movement Datasets" section), most of them are not explicitly annotated with ground truth movement patterns. Moreover, despite the growing availability of tools for movement data processing and analysis, there remain few open-source implementations of pattern mining algorithms, especially for complex patterns such as swarms, traveling companions, and convergence/divergence. Future efforts should focus on creating and sharing high-quality benchmark datasets and algorithm repositories. These benchmarks should cover diverse scenarios, moving object types, and movement pattern categories to support standardized preprocessing, comprehensive evaluation, and scalable implementations.

Deep Learning for Movement Pattern Mining

In recent years, traditional data mining methods (e.g., clustering and template matching) have dominated movement pattern mining. With the continuous improvement of data volume and computing power, deep learning has flourished in many application fields and achieved remarkable results. Integrating deep learning techniques into movement pattern mining is a valuable but challenging research frontier. It is necessary to solve movement data processing and representation, multi-dimensional feature learning, and inter-object relationship modeling, improve the generalization ability of deep learning models, and ensure efficient performance in real-time scenarios. Meanwhile, the lack of standard movement pattern dataset benchmarks and open-source implementation of deep learning-based pattern mining models presents a significant challenge. Future efforts should also focus on developing and disseminating high-quality benchmark datasets and code repositories to accelerate progress in this field.

Evaluation and Interpretation

Comprehensive evaluation and interpretation of movement pattern mining results are essential for practical applications. Evaluation includes both qualitative assessment through visualization and quantitative evaluation using appropriate metrics. However, designing effective visualization techniques to accurately convey the extracted movement patterns and developing robust quantitative metrics to assess accuracy and reliability is challenging. Future research should focus on domain-specific visualization tools and metrics that can evaluate the performance of different aspects of the detected movement patterns.

Interpretation, however, focuses on explaining the underlying semantics and potential causality of the discovered movement patterns. Compared to existing varieties of trajectory semantic mining, research on semantic association and interpretation of high-level movement patterns remains relatively limited. Further attention is needed to explore how contextual and environmental factors can be integrated to enhance the understanding of movement behaviors of moving objects in diverse real-world scenarios.

Table 2 The comparison of accompanying movement patterns

No.	Pattern type		Fixed disc size	Stationary area	Density-connected	Time gaps	Persistent members	Constant members	Absence at certain times	Group size change	Minimum duration	Minimum object size	Local continuity
1	Flock	Fixed/sStandard flock	√				√	√			√	√	
		Varying flock	√				√		√	√	√	√	
		Static flock	√	√			√	√			√	√	
		Moving flock	√				√	√			√	√	
2	Convoy	Standard convoy			√		√	√			√	√	
		Evolving convoy			√		√			√	√	√	
		Dynamic convoy			√		√	√	√	√	√	√	
3	Moving cluster				√				√	√	√	√	
4	Swarm	Standard swarm			√	√	√	√			√	√	
		Fuzzy swarm			√	√	√	√			√	√	
5	Traveling companion	Standard companion			√		√	√			√	√	
		Loose traveling companion			√	√	√	√		√	√	√	
		Loose group companion			√		√		√	√	√	√	
6	Platoon				√	√	√	√	√	√	√	√	√
7	Gathering				√		√		√	√	√	√	
8	Real-Gpattern	Real-Gpattern +			√	√			√	√	√	√	
		Real-Gpattern-			√	√			√	√	√	√	√
9	Snowball	Snowball +			√		√			√	√	√	√
		Snowball-			√					√	√	√	
10	Leader-followership	Fixed followers	√				√	√		√	√	√	
		Varying followers	√							√	√	√	

Privacy and Ethical Considerations in Movement Data Mining

Privacy and ethics concerns are critical throughout the collection, mining, and analysis of movement data related to humans. Movement data may contain sensitive personal details. Research has shown that a few spatiotemporal points can uniquely identify most individuals (De Montjoye et al. 2013). Meanwhile, with advanced analytics and artificial intelligence methods, the ability to extract sensitive information (e.g., social relationships and personal habits) from movement data has increased. This amplifies the ethical responsibility of researchers and promotes privacy-preserving techniques (e.g., differential privacy, k-anonymity, synthetic data generation) to be urgently considered. In the human-centered era, ethical considerations should precede technical safeguards. The purpose of collecting movement data, the consent of individuals involved, and potential misuse are all of high concern. Future research must balance innovation in pattern mining with responsible data governance and adherence to ethical standards.

Conclusions

Movement pattern mining holds significant theoretical and practical value across various domains and disciplines. This study presents a comprehensive review of the current landscape of movement pattern studies, aiming to provide a systematic overview of progress in this field. First, we clarify the basic concepts in movement pattern mining, which are essential for readers to establish a common foundation. We then introduce movement data characteristics, as they are necessary for the design of effective pattern-mining algorithms. Regarding the existing various types of movement patterns, we summarize eight major categories and provide formal definitions and visualizations for each pattern and its multiple variants.

Furthermore, we synthesize ten representative movement pattern mining algorithms, including traditional data mining techniques and emerging deep learning models. To highlight real-world applicability, we present several typical applications such as transportation, human behaviors, ecology, and maritime monitoring. We also discuss challenges of existing movement pattern research, which need to be further addressed and explored, such as scale-adapted movement pattern recognition, coupled with multiple movement pattern mining, and the integration of deep learning models.

Although this study offers a broad overview of movement pattern research, some emerging or rare movement patterns may still be overlooked. In addition, due to the diversity of pattern types and detection algorithms, this study does not provide empirical benchmarks or performance comparisons of various movement pattern mining algorithms. Future research should focus on developing standardized movement datasets and open-source tools to enhance the transparency and reproducibility of movement pattern studies. It is also vital and of great potential to investigate the integration of deep learning models with interpretable pattern mining and to simultaneously address the detection of multiple movement patterns. Moreover, online movement pattern detection remains challenging and promising, especially for real-time applications such as traffic monitoring, emergency response, and surveillance.

Appendix

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References

Agrawal R, Srikant R (1994) Fast algorithms for mining association rules. Proceedings of the 20th International Conference on Very Large Data Bases, Santiago, Chile. ACM, pp 478–499

Agrawal R, Srikant R (1995) Mining sequential patterns. Proceedings of the eleventh international conference on data engineering. IEEE, pp 3–14

Al-Ajlan A (2015) The comparison between forward and backward chaining. Int J Mach Learn Comput 5(2):106

Alamri S, Taniar D, Safar M (2014) A taxonomy for moving object queries in spatial databases. Future Gener Comput Syst 37:232–242

Andersson M, Gudmundsson J, Laube P et al (2007) Reporting leadership patterns among trajectories. Proceedings of the ACM Symposium on Applied Computing, pp 3–7

Andersson M, Gudmundsson J, Laube P et al (2008) Reporting leaders and followers among trajectories of moving point objects. GeoInformatica 12:497–528

Andrienko N, Andrienko G (2007) Designing visual analytics methods for massive collections of movement data. Cartographica 42(2):117–138

Andrienko N, Andrienko G (2013) Visual analytics of movement: an overview of methods, tools and procedures. Inf vis 12(1):3–24

Andrienko N, Andrienko G, Miksch S et al (2021) A theoretical model for pattern discovery in visual analytics. Vis Inf 5(1):23–42

Ansari MY, Ahmad A, Khan SS et al (2020) Spatiotemporal clustering: a review. Artif Intell Rev 53:2381–2423

Arentze TA, Hofman F, Van Mourik H et al (2000) Using decision tree induction systems for modeling space-time behavior. Geogr Anal 32(4):330–350

Atluri G, Karpatne A, Kumar V (2018) Spatio-temporal data mining: a survey of problems and methods. ACM Comput Surv 51(4):1–41

Aung HH, Tan KL (2010) Discovery of evolving convoys. Scientific and Statistical Database Management: 22nd International Conference, SSDBM (2010) Heidelberg, Germany, June 30–July 2, 2010. Proceedings 22. Springer, Berlin Heidelberg, pp 196–213

Ayres J, Flannick J, Gehrke J et al (2002) Sequential pattern mining using a bitmap representation. Proceedings of the eighth ACM SIGKDD international conference on Knowledge discovery and data mining, pp 429–435

Baglioni M, Fernandes de Macêdo J A, Renso C et al (2009) Towards semantic interpretation of movement behavior. Advances in GIScience: Proceedings of the 12th AGILE Conference. Berlin, Heidelberg: Springer Berlin Heidelberg, pp 271–288

Bareche I, Xia Y (2022) A distributed hybrid indexing for continuous KNN query processing over moving objects. ISPRS Int J Geo-Inf 11(4):264

Behrisch M, Bieker L, Erdmann J et al (2011) SUMO–simulation of urban mobility: an overview. Proceedings of SIMUL 2011, The Third International Conference on Advances in System Simulation, pp 55–60

Benkert M, Gudmundsson J, Hübner F et al (2008) Reporting flock patterns. Comput Geom 41(3):111–125

Bennison A, Bearhop S, Bodey TW et al (2018) Search and foraging behaviors from movement data: a comparison of methods. Ecol Evol 8(1):13–24

Berndt DJ, Clifford J (1994) Using dynamic time warping to find patterns in time series. Proceedings of the 3rd International Conference on Knowledge Discovery and Data Mining, pp 359–370

Bhattacharjee P, Mitra P (2021) A survey of density based clustering algorithms. Front Comput Sci 15:1–27

Bogorny V, Kuijpers B, Alvares LO (2009) St-dmql: a semantic trajectory data mining query language. Int J Geogr Inf Sci 23(10):1245–1276

Borah A, Nath B (2018) Fp-tree and its variants: towards solving the pattern mining challenges. Proceedings of First International Conference on Smart System, Innovations and Computing: SSIC 2017, Jaipur, India. Springer Singapore, pp 535–543

Breitenbach T, Wilkusz B, Rasbach L et al (2023) On a method for detecting periods and repeating patterns in time series data with autocorrelation and function approximation. Pattern Recogn 138:109355

Brinkhoff T (2002) A framework for generating network-based moving objects. GeoInformatica 6(2):153–180

Cao H, Mamoulis N, Cheung DW (2007) Discovery of periodic patterns in spatiotemporal sequences. IEEE Trans Knowl Data Eng 19(4):453–467

Cao N, Lin C, Zhu Q et al (2017) Voila: Visual anomaly detection and monitoring with streaming spatiotemporal data. IEEE Trans Visual Comput Graphics 24(1):23–33

Chanda AK, Saha S, Nishi MA et al (2015) An efficient approach to mine flexible periodic patterns in time series databases. Eng Appl Artif Intell 44:46–63

Chang Y, Wang X, Wang J et al (2024) A survey on evaluation of large language models. ACM Trans Intell Syst Technol 15(3):1–45

Chaudhry R, Hager G, Vidal R (2013) Dynamic template tracking and recognition. Int J Comput Vis 105:19–48

Chen JS, Wang W, Prebensen NK (2016) Travel companions and activity preferences of nature-based tourists. Tour Rev 71(1):45–56

Chen L, Gao Y, Fang Z et al (2019) Real-time distributed co-movement pattern detection on streaming trajectories. Proc VLDB Endow 12(10):1208–1220

Chen L, Özsu MT, Oria V (2005) Robust and fast similarity search for moving object trajectories. Proceedings of the. ACM SIGMOD International Conference on Management of Data 2005:491–502

Chen P, Liu S, Lukač N (2023) CNN-based vessel meeting knowledge discovery from AIS vessel trajectories. J Database Manag 34(3):1–38

Cheng R, Kao B, Kwan A et al (2009) Filtering data streams for entity-based continuous queries. IEEE Trans Knowl Data Eng 22(2):234–248

Chu C, Zhang H, Wang P et al (2024) Simulating human mobility with a trajectory generation framework based on diffusion model. Int J Geogr Inf Sci 38(5):847–878

Cordeiro M, Sarmento RP, Gama J (2016) Dynamic community detection in evolving networks using locality modularity optimization. Soc Netw Anal Min 6:1–20

Da Silva TLC, de Macêdo JAF, Casanova MA (2014) Discovering frequent mobility patterns on moving object data. Proceedings of the Third ACM SIGSPATIAL International Workshop on Mobile Geographic Information Systems, pp 60–67

de Lucca Siqueira F, Bogorny V (2011) Discovering chasing behavior in moving object trajectories. Trans GIS 15(5):667–688

De Montjoye YA, Hidalgo CA, Verleysen M et al (2013) Unique in the crowd: the privacy bounds of human mobility. Sci Rep 3(1):1376

Demšar U, Buchin K, Cagnacci F et al (2015) Analysis and visualisation of movement: an interdisciplinary review. Mov Ecol 3:1–24

Demšar U, Long JA, Benitez-Paez F et al (2021) Establishing the integrated science of movement: bringing together concepts and methods from animal and human movement analysis. Int J Geogr Inf Sci 35(7):1273–1308

Deng M, Shi Y, Gong J et al (2016) A summary of spatio-temporal outlier detection. Geogr Geo-Inform Sci 32(06):43–50

Devall B (1980) The deep ecology movement. Nat Resources J 20:299

Dodge S (2021) A data science framework for movement. Geogr Anal 53(1):92–112

Dodge S, Weibel R, Lautenschütz AK (2008) Towards a taxonomy of movement patterns. Inf Vis 7(3–4):240–252

Dodge S, Weibel R, Forootan E (2009) Revealing the physics of movement: comparing the similarity of movement characteristics of different types of moving objects. Comput Environ Urban Syst 33(6):419–434

Dodge S, Weibel R, Ahearn SC et al (2016) Analysis of movement data. Int J Geogr Inf Sci 30(5):825–834

Dodge S, Gao S, Tomko M et al (2020) Progress in computational movement analysis–towards movement data science. Int J Geogr Inf Sci 34(12):2395–2400

Dong X, Pi DC (2013) Novel method for hurricane trajectory prediction based on data mining. Nat Hazards Earth Syst Sci 13(12):3211–3220

Düntgen C, Behr T, Güting RH (2009) BerlinMOD: a benchmark for moving object databases. VLDB J 18:1335–1368

Edelhoff H, Signer J, Balkenhol N (2016) Path segmentation for beginners: an overview of current methods for detecting changes in animal movement patterns. Mov Ecol 4:1–21

Eiter T, Mannila H (1994) Computing discrete Fréchet distance. Technische Universitat Wien Technical report
Ester M, Kriegel H P, Sander J et al (1996) A density-based algorithm for discovering clusters in large spatial databases with noise. KDD 96(34):226–231
Evans W, Kirkpatrick D, Loffler M et al (2016) Minimizing co-location potential of moving entities. SIAM J Comput 45(5):1870–1893
Fang Z, Yang X, Xu Y et al (2017) Spatiotemporal model for assessing the stability of urban human convergence and divergence patterns. Int J Geogr Inf Sci 31(11):2119–2141
Farahnakian F, Nicolas F, Farahnakian F et al (2023) A comprehensive study of clustering-based techniques for detecting abnormal vessel behavior. Remote Sens 15(6):1477
Fileto R, Bogorny V, May C et al (2015) Semantic enrichment and analysis of movement data: probably it is just starting! SIGSPATIAL Special 7(1):11–18
Fisher PF, Laube P, van Kreveld M et al (2005) Finding REMO—detecting relative motion patterns in geospatial lifelines. Developments in Spatial Data Handling: 11th International Symposium on Spatial Data Handling. Springer, Berlin Heidelberg, pp 201–215
Fister I Jr, Fister I, Fister D et al (2023) A comprehensive review of visualization methods for association rule mining: taxonomy, challenges, open problems and future ideas. Expert Syst Appl 233:120901
Forgy CL (1989) Rete: a fast algorithm for the many pattern/many object pattern match problem. In: Mylopolous J, Brodie M (eds) Readings in artificial intelligence and databases. Morgan Kaufmann, San Francisco, pp 547–559
Fournier-Viger P, Lin JCW, Kiran RU et al (2017) A survey of sequential pattern mining. Data Sci Pattern Recognit 1(1):54–77
Fournier-Viger P, Yang P, Kiran RU et al (2021) Mining local periodic patterns in a discrete sequence. Inf Sci 544:519–548
Francia M, Gallinucci E, Golfarelli M (2024) Colossal trajectory mining: a unifying approach to mine behavioral mobility patterns. Expert Syst Appl 238:122055
Frank J, Mannor S, Pineau J et al (2012) Time series analysis using geometric template matching. IEEE Trans Pattern Anal Mach Intell 35(3):740–754
Fu T, Chung F, Luk R et al (2007) Stock time series pattern matching: template-based vs. rule-based approaches. Eng Appl Artif Intell 20(3):347–364
Furletti B, Cintia P, Renso C et al (2013) Inferring human activities from GPS tracks. Proceedings of the 2nd ACM SIGKDD international workshop on urban computing, pp 1–8
Gahtan J, Knapp KR, Schreck CJ et al (2024) International best track archive for climate stewardship (IBTrACS) Project, Version 4r01, NOAA National Centers for Environmental Information[EB/OL]
Gan W, Lin JCW, Fournier-Viger P et al (2019) A survey of parallel sequential pattern mining. ACM Trans Know Discov Data (TKDD) 13(3):1–34
Gao X, Xiao B, Tao D et al (2010) A survey of graph edit distance. Pattern Anal Appl 13:113–129
Gao Y, Huang L, Feng J et al (2020) Semantic trajectory segmentation based on change-point detection and ontology. Int J Geogr Inf Sci 34(12):2361–2394
Gonzalez MC, Hidalgo CA, Barabasi AL (2008) Understanding individual human mobility patterns. Nature 453(7196):779–782
Graser A (2019) Movingpandas: efficient structures for movement data in python. Giforum 1:54–68
Graser A, Dragaschnig M, Koller H (2020) Exploratory analysis of massive movement data. Handbook of big geospatial data. Springer International Publishing, Cham, pp 285–319
Gudmundsson J, Van Kreveld M (2006) Computing longest duration flocks in trajectory data. Proceedings of the 14th Annual ACM international symposium on advances in geographic information systems pp, 35–42
Guibas LJ, Roeloffzen M (2017) Modeling motion. In: Goodman JE, O'Rourke J, Tóth CS (eds) Handbook of Discrete and Computational Geometry, 3rd edn. Chapman and Hall/CRC, Boca Raton, pp 1401–1420
Guo L, Huang G, Ding Z (2014) fficient detection of emergency event from moving object data streams. Database Systems for Advanced Applications: 19th International Conference, DASFAA (2014) Bali, Indonesia, April 21–24, 2014. Proceedings, Part II 19. Springer International Publishing, pp 422–437
Hai PN, Ienco D, Poncelet P et al (2012b) Mining time relaxed gradual moving object clusters. Proceedings of the 20th international conference on advances in geographic information systems, pp 478–481
Hai PN, Ienco D, Poncelet P et al (2012a) Mining fuzzy moving object clusters. Advanced Data Mining and Applications: 8th International Conference, ADMA 2012, Nanjing, China, December 15–18, 2012. Proceedings 8. Springer Berlin Heidelberg, pp 100–114
Han J, Pei J (2000) Mining frequent patterns by pattern-growth: methodology and implications. ACM SIGKDD Explor Newsl 2(2):14–20
Han J, Pei J, Mortazavi-Asl B et al (2000b) FreeSpan: frequent pattern-projected sequential pattern mining. Proceedings of the Sixth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp 355–359
Han J, Pei J, Mortazavi-Asl B et al (2001) Prefixspan: mining sequential patterns efficiently by prefix-projected pattern growth. Proceedings of the 17th International Conference on Data Engineering. Piscataway, NJ, USA: IEEE pp 215–224
Han J, Pei J, Yin Y et al (2004) Mining frequent patterns without candidate generation: a frequent-pattern tree approach. Data Min Knowl Discov 8:53–87
Hassan SM, Al-Sadek AF, Hemayed EE (2010) Rule-based approach for enhancing the motion trajectories in human activity recognition. 2010 10th International Conference on Intelligent Systems Design and Applications. IEEE, pp 829–834
He J, Chen H, Chen Y et al (2019) Diverse visualization techniques and methods of moving-object-trajectory data: a review. ISPRS Int J Geo-Inf 8(2):63
He Q, Chen Y, Dong Q et al (2020) Mining moving object gathering pattern based on resilient distributed datasets and R-tree index. Neurocomputing 393:194–202
Helmi S, Banaei-Kashani F (2017) Spatiotemporal range pattern queries on large-scale co-movement pattern datasets. 2017 IEEE International Conference on Big Data (Big Data). IEEE, pp 425–434
Heupel MR, Munroe SEM, Lédée EJI et al (2019) Interspecific interactions, movement patterns and habitat use in a diverse coastal shark assemblage. Mar Biol 166:1–17
Hindmarsh GP, Black AA, White SLJ et al (2021) Eye movement patterns and reading ability in children. Ophthalmic Physiol Opt 41(5):1134–1143
Holyoak M, Casagrandi R, Nathan R et al (2008) Trends and missing parts in the study of movement ecology. Proc Natl Acad Sci U S A 105(49):19060–19065
Huang Z, Gedeon TD, Nikravesh M (2008) Pattern trees induction: a new machine learning method. IEEE Trans Fuzzy Syst 16(4):958–970
Jain AK, Duin RPW, Mao J (2000) Statistical pattern recognition: a review. IEEE Trans Pattern Anal Mach Intell 22(1):4–37
Jeung H, Yiu M L, Zhou X et al (2008) Discovery of convoys in trajectory databases. Proceedings of the VLDB Endowment 1:1068–1080. https://doi.org/10.14778/1453856.145397

Ji G, Sun H, Zhao B (2016) Research progress in group moving patterns mining of spatio-temporal trajectories. J Nanjing Univ Aeronaut Aeronaut 48(5):615–624

Jia J, Hu Y, Zhao B et al (2021) Discovering collective converging groups of large scale moving objects in road networks[C]// Database Systems for Advanced Applications: 26th International Conference, DASFAA 2021, Taipei, Taiwan, April 11–14, 2021, Proceedings, Part II 26. Springer International Publishing, pp 307–324

Jia J, Ji G, Zhao B (2025) Mining converging patterns over streaming trajectories of moving objects in road networks. Knowledge-Based Systems 309:112883

Joo R, Picardi S, Boone ME et al (2022) Recent trends in movement ecology of animals and human mobility. Mov Ecol 10(1):26

Kalnis P, Mamoulis N, Bakiras S (2005) On discovering moving clusters in spatio-temporal data. Advances in Spatial and Temporal Databases: 9th International Symposium, SSTD (2005) Angra dos Reis, Brazil, August 22–24, 2005. Proceedings 9. Springer, Berlin Heidelberg, pp 364–381

Kapp A, Hansmeyer J, Mihaljević H (2023) Generative models for synthetic urban mobility data: a systematic literature review. ACM Comput Surv 56(4):1–37

Kays R, Davidson SC, Berger M et al (2022) The Movebank system for studying global animal movement and demography. Methods Ecol Evol 13(2):419–431

Kays R, Hirsch B, Caillaud D et al (2023) Multi-scale movement syndromes for comparative analyses of animal movement patterns. Mov Ecol 11(1):61

Khazukov K, Shepelev V, Karpeta T et al (2020) Real-time monitoring of traffic parameters. J Big Data 7:1–20

Kim SS, Song KS, Go JH (2018) A Study of design with spatial rule-based engine using geospatial big-data. 2018 International Conference on Information and Communication Technology Convergence (ICTC). IEEE, pp 1273–1275

Kim H, Yun U, Vo B et al (2020) Periodicity-oriented data analytics on time-series data for intelligence system. IEEE Syst J 15(4):4958–4969

Kleanthous N, Hussain AJ, Khan W et al (2022) A survey of machine learning approaches in animal behaviour. Neurocomputing 491:442–463

Knapp KR, Kruk MC, Levinson DH et al (2010) The international best track archive for climate stewardship (IBTrACS) unifying tropical cyclone data. Bull Am Meteorol Soc 91(3):363–376

Kontopoulos I, Varlamis I, Tserpes K (2021) A distributed framework for extracting maritime traffic patterns. Int J Geogr Inf Sci 35(4):767–792

Kumbhare TA, Chobe SV (2014) An overview of association rule mining algorithms. Int J Comput Sci Inf Technol 5(1):927–930

Landsea CW, Franklin JL (2013) Atlantic hurricane database uncertainty and presentation of a new database format. Mon Weather Rev 141(10):3576–3592

Laube P (2014) Computational movement analysis. Springer International Publishing, Cham, pp 31–32

Laube P, Imfeld S (2002) Analyzing relative motion within groups oftrackable moving point objects. International Conference on Geographic Information Science. Berlin, Heidelberg: Springer Berlin Heidelberg, pp 132–144

Laube P, Imfeld S, Weibel R (2005) Discovering relative motion patterns in groups of moving point objects. Int J Geogr Inf Sci 19(6):639–668

Lee JG, Han J, Whang KY (2007) Trajectory clustering: a partition-and-group framework. Proceedings of the. ACM SIGMOD Int Conf Manag Data 593–604

Lehmann AL, Alvares LO, Bogorny V (2019) SMSM: a similarity measure for trajectory stops and moves. Int J Geogr Inf Sci 33(9):1847–1872

Li Z (2014) Spatiotemporal pattern mining: algorithms and applications. Frequent pattern mining 283–306

Li Z, Ding B, Han J, et al (2010b) Mining periodic behaviors for moving objects[C]//Proceedings of the 16th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining :1099–1108

Li Y, Han J, Yang J (2004) Clustering moving objects. Proceedings of the tenth ACM SIGKDD Int Conf Knowl Discov Data Min, pp 617–622

Li Z, Ding B, Han J et al (2010) Swarm: mining relaxed temporal moving object clusters. Proc VLDB Endow 3(1–2):723–734

Li Z, Han J, Ji M et al (2011) Movemine: mining moving object data for discovery of animal movement patterns. ACM Trans Intell Syst Technol 2(4):1–32

Li Z, Han J, Ding B et al (2012) Mining periodic behaviors of object movements for animal and biological sustainability studies. Data Min Knowl Discov 24:355–386

Li Y, Bailey J, Kulik L (2015) Efficient mining of platoon patterns in trajectory databases. Data Knowl Eng 100:167–187

Li J, Wang J, Zhang J et al (2016) A probabilistic approach to detect mixed periodic patterns from moving object data. GeoInformatica 20:715–739

Li K, Wang H, Chen Z et al (2023) Relaxed group pattern detection over massive-scale trajectories. Future Gener Comput Syst 144:131–139

Li X, Zhen X, Qi X et al (2023) Dynamic community detection based on graph convolutional networks and contrastive learning. Chaos Solitons Fractals 176:114157

Liang G, Kintak U, Ning X et al (2023) Semantics-aware dynamic graph convolutional network for traffic flow forecasting. IEEE Trans Veh Technol 72(6):7796–7809

Lim GJ, Zangeneh S, Kim SJ (2016) Clustering approach for defining hurricane evacuation zones. J Urban Plann Dev 142(4):04016008

Liu C, Guo C (2020) STCCD: semantic trajectory clustering based on community detection in networks. Expert Syst Appl 162:113689

Liu Y, Zhao Y, Chen L et al (2011) Mining frequent trajectory patterns for activity monitoring using radio frequency tag arrays. IEEE Trans Parallel Distrib Syst 23(11):2138–2149

Liu M, He G, Long Y (2021) A semantics-based trajectory segmentation simplification method. Journal of Geovisualization and Spatial Analysis 5(2):19

Liu Y, Dai H, Li J et al (2023) BP-model-based convoy mining algorithms for moving objects. Expert Syst Appl 213:118860

Long JA (2016) Kinematic interpolation of movement data. Int J Geogr Inf Sci 30(5):854–868

Ma X, Huo E, Yu H et al (2021) Mining truck platooning patterns through massive trajectory data. Knowledge-Based Systems 221:106972

Mabroukeh NR, Ezeife CI (2010) A taxonomy of sequential pattern mining algorithms. ACM Comput Surv 43(1):1–41

Madhulatha TS (2012) An overview on clustering methods. IOSR Journal of Engineering. https://doi.org/10.9790/3021-0204719725

Maiti S, Subramanyam RBV (2021) Mining co-location patterns from distributed spatial data. J King Saud Univ-Comput Inf Sci 33(9):1064–1073

Martin H, Hong Y, Wiedemann N et al (2023) Trackintel: an open-source Python library for human mobility analysis. Comput Environ Urban Syst 101:101938

May Petry L, Leite Da Silva C, Esuli A et al (2020) MARC: a robust method for multiple-aspect trajectory classification via space, time, and semantic embeddings. International Journal of Geographical Information Science 34(7):1428–1450

Mazumdar N, Sarma PKD (2024) Sequential pattern mining algorithms and their applications: a technical review. Int J Data Sci Anal 1–44

Michalski RS (1980) Pattern recognition as rule-guided inductive inference. IEEE Trans Pattern Anal Mach Intell 4:349–361
Miller HJ, Dodge S, Miller J et al (2019) Towards an integrated science of movement: converging research on animal movement ecology and human mobility science. Int J Geogr Inf Sci 33(5):855–876
Moreira-Matias L, Gama J, Ferreira M et al (2013) Predicting taxi–passenger demand using streaming data. IEEE Trans Intell Transp Syst 14(3):1393–1402
Morris B, Trivedi M (2009) Learning trajectory patterns by clustering: experimental studies and comparative evaluation. 2009 IEEE Conference on Computer Vision and Pattern Recognition. IEEE, 312–319
Mueen A (2014) Time series motif discovery: dimensions and applications. Wires Data Min Knowl Discov 4(2):152–159
Nanni M, Pedreschi D (2006) Time-focused clustering of trajectories of moving objects. J Intell Inf Syst 27:267–289
Naserian E, Wang X, Xu X et al (2018) A framework of loose travelling companion discovery from human trajectories. IEEE Trans Mob Comput 17(11):2497–2511
Nathan R, Getz WM, Revilla E et al (2008) A movement ecology paradigm for unifying organismal movement research. Proc Natl Acad Sci U S A 105(49):19052–19059
Nathan R, Monk CT, Arlinghaus R et al (2022) Big-data approaches lead to an increased understanding of the ecology of animal movement. Science 375(6582):eabg1780
Nishi MA, Ahmed CF, Samiullah M et al (2013) Effective periodic pattern mining in time series databases. Expert Syst Appl 40(8):3015–3027
Niyazmand T, Izadi I (2019) Pattern mining in alarm flood sequences using a modified PrefixSpan algorithm. ISA Trans 90:287–293
Okubo A (1986) Dynamical aspects of animal grouping: swarms, schools, flocks, and herds. Adv Biophys 22:1–94
Orakzai F, Pedersen TB, Calders T (2021) Distributed mining of convoys in large scale datasets. GeoInformatica 25(2):353–396
Pan Z, Liu P, Yi J (2018) An improved FP-tree algorithm for mining maximal frequent patterns. Proceedings of the 2018 10th International Conference on Measuring Technology and Mechatronics Automation (ICMTMA), pp 309–312
Pappalardo L, Simini F (2018) Data-driven generation of spatio-temporal routines in human mobility. Data Min Knowl Discov 32(3):787–829
Pappalardo L, Simini F, Barlacchi G et al (2022) Scikit-mobility: a python library for the analysis, generation, and risk assessment of mobility data. J Stat Softw 103:1–38
Parent C, Spaccapietra S, Renso C et al (2013) Semantic trajectories modeling and analysis. ACM Comput Surv 45(4):1–32
Peng J, Tang J, Deng M et al (2025) A two-stage method for detecting trajectory clusters of different densities with peak trajectories identification. Int J Geogr Inf Sci 1–34
Peng J, Liu H, Tang J et al (2023) Exploring crowd travel demands based on the characteristics of spatiotemporal interaction between urban functional zones. ISPRS Int J Geo-Inf 12(6):225
Peng J, Deng M, Tang J et al (2024) A movement-aware measure for trajectory similarity and its application for ride-sharing path extraction in a road network. Int J Geogr Inf Sci 38(9):1703–1727
Pfoser D, Jensen CS (1999) Capturing the uncertainty of moving-object representations. International symposium on spatial databases. Berlin, Heidelberg: Springer Berlin Heidelberg, pp 111–131
Puntheeranurak S, Shein TT, Imamura M (2018) Efficient discovery of traveling companion from evolving trajectory data stream. 2018 IEEE 42nd Annual Computer Software and Applications Conference (COMPSAC). IEEE 1:448–453
Qian T, Sun S, Shan X et al (2022) Distributed-swarm: a real-time pattern detection model based on density clustering. IEEE Access 10:59832–59842
Qin W, Tang J, Lu C et al (2021) A trajectory abnormal detection method based on segmentation and clustering. J Phy Conf Ser. IOP Publishing 2010(1):012188
Raiyn J, Toledo T (2014) Real-time road traffic anomaly detection. J Transp Technol 4(3):256–266
Rasheed F, Alhajj R (2010) STNR: a suffix tree based noise resilient algorithm for periodicity detection in time series databases. Appl Intell 32:267–278
Renso C, Baglioni M, de Macedo JAF et al (2013) How you move reveals who you are: understanding human behavior by analyzing trajectory data. Knowl Inf Syst 37:331–362
Saraf P, Sedamkar R, Rathi S (2015) Prefixspan algorithm for finding sequential pattern with various constraints. Int J Appl Inf Syst 9(3):37–41
Saxena A, Prasad M, Gupta A et al (2017) A review of clustering techniques and developments. Neurocomputing 267:664–681
Seidel DP, Dougherty ER, Getz WM (2019) Exploratory movement analysis and report building with R package stmove. bioRxiv 758987. https://doi.org/10.1101/758987
Senge R, Hüllermeier E (2010) Top-down induction of fuzzy pattern trees. IEEE Trans Fuzzy Syst 19(2):241–252
Shaker A, Senge R, Hüllermeier E (2013) Evolving fuzzy pattern trees for binary classification on data streams. Inf Sci 220:34–45
Shanthappa NK, Mulangi RH, Manjunath HM (2023) The spatiotemporal patterns of bus passengers: visualisation and evaluation using non-negative tensor decomposition. Journal of Geovisualization and Spatial Analysis 7(1):9
Shaw AA, Gopalan NP (2014) Finding frequent trajectories by clustering and sequential pattern mining. Journal of Traffic and Transportation Engineering (English Edition) 1(6):393–403
Shein TT, Puntheeranurak S, Imamura M (2020) Discovery of loose group companion from trajectory data streams. IEEE Access 8:85856–85868
Shein TT, Puntheeranurak S, Imamura M (2020) Discovery of evolving companion from trajectory data streams. Knowl Inf Syst 62(9):3509–3533
Shenk J, Byttner W, Nambusubramaniyan S et al (2021) Traja: a Python toolbox for animal trajectory analysis. J Open Source Softw 6(63):3202
Shi Y, Long C, Yang X et al (2022) Abnormal ship behavior detection based on AIS data. Appl Sci 12(9):4635
Shi Y, Chen Y, Deng M et al (2022) Discovering source areas of disease outbreaks based on ring-shaped hotspot detection in road network space. Int J Geogr Inf Sci 36(7):1343–1363
Sirisha G, Shashi M, Raju GVP (2013) Periodic pattern mining–algorithms and applications. Global Journal of Computer Science and Technology Software & Data Engineering 13(13):19–28
Soleimani BH, De Souza EN, Hilliard C et al (2015) Anomaly detection in maritime data based on geometrical analysis of trajectories. 2015 18th International Conference on Information Fusion (Fusion). IEEE 1100–1105
Soni H K, Sharma S, Jain M (2016) Frequent pattern generation algorithms for association rule mining: strength and challenges. Proceedings of the 2016 International Conference on Electrical, Electronics, and Optimization Techniques (ICEEOT), pp 3744–3747
Srikant R, Agrawal R (1996) Mining sequential patterns: generalizations and performance improvements. International Conference on Extending Database Technology. Berlin, Heidelberg: Springer Berlin Heidelberg 1–17
Stergiopoulos G, Valvis E, Mitrodimas D et al (2018) Analyzing congestion interdependencies of ports and container ship

routes in the maritime network infrastructure. IEEE Access 6:63823–63832
Su R, McBride EC, Goulias KG (2020) Pattern recognition of daily activity patterns using human mobility motifs and sequence analysis. Transportation Research Part c: Emerging Technologies 120:102796
Sweeting AJ, Aughey RJ, Cormack SJ et al (2017) Discovering frequently recurring movement sequences in team-sport athlete spatiotemporal data. J Sports Sci 35(24):2439–2445
Tanaka PS, Vieira MR, Kaster DS (2016) An improved base algorithm for online discovery of flock patterns in trajectories. J Inf Data Manag 7(1):52–52
Tang L A, Zheng Y, Yuan J et al (2012) On discovery of traveling companions from streaming trajectories. 2012 IEEE 28th International Conference on Data Engineering. IEEE 186–197
Tang LA, Zheng Y, Yuan J et al (2014) A framework of traveling companion discovery on trajectory data streams. ACM Trans Intell Syst Technol 5(1):1–34
Trajcevski G, Wolfson O, Hinrichs K et al (2004) Managing uncertainty in moving objects databases. ACM Trans Database Syst 29(3):463–507
Tritsarolis A, Theodoropoulos GS, Theodoridis Y (2021) Online discovery of co-movement patterns in mobility data. Int J Geogr Inf Sci 35(4):819–845
Tritsarolis A, Chondrodima E, Tampakis P et al (2024) Predicting co-movement patterns in mobility data. GeoInformatica 28(2):221–243
Tsai HP, Yang DN, Chen MS (2009) Mining group movement patterns for tracking moving objects efficiently. IEEE Trans Knowl Data Eng 23(2):266–281
Tu E, Zhang G, Mao S et al (2020) Modeling historical AIS data for vessel path prediction: a comprehensive treatment. arXiv preprint arXiv:2001.01592
Turdukulov U, Calderon Romero AO, Huisman O et al (2014) Visual mining of moving flock patterns in large spatio-temporal data sets using a frequent pattern approach. Int J Geogr Inf Sci 28(10):2013–2029
Van der Kruk E, Reijne MM (2018) Accuracy of human motion capture systems for sport applications; state-of-the-art review. Eur J Sport Sci 18(6):806–819
Vandecasteele A, Devillers R, Napoli A (2014) From movement data to objects behavior using semantic trajectory and semantic events. Mar Geod 37(2):126–144
Vieira MR, Bakalov P, Tsotras VJ (2009) On-line discovery of flock patterns in spatio-temporal data. Proceedings of the 17th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems 286–295
Wachowicz M, Ong R, Renso C et al (2011) Finding moving flock patterns among pedestrians through collective coherence. Int J Geogr Inf Sci 25(11):1849–1864
Wang S, Wu L, Zhou F et al (2015) Group pattern mining on moving objects' uncertain trajectories. Int J Comput Commun Control 10(3):428–440
Wang S, Cao J, Philip SY (2020) Deep learning for spatio-temporal data mining: a survey. IEEE Trans Knowl Data Eng 34(8):3681–3700
Wang J, Liang Y, Tang J et al (2024) Vehicle trajectory reconstruction using lagrange-interpolation-based framework. Appl Sci 14(3):1173
Wen R, Yan W (2019) Vessel crowd movement pattern mining for maritime traffic management. LOGI – Scientific Journal on Transport and Logistics 10(2):105–115
Wu F, Lei TKH, Li Z et al (2014) Movemine 2.0: mining object relationships from movement data. Proc VLDB Endow 7(13):1613–1616
Wu H, Tang X, Wang Z et al (2021) Uncovering abnormal behavior patterns from mobility trajectories. Sensors (Basel) 21(10):3520
Yadamjav ME, Bao Z, Zheng B et al (2020) Querying recurrent convoys over trajectory data. ACM Trans Intell Syst Technol 11(5):1–24
Yan Z, Chakraborty D, Parent C et al (2013) Semantic trajectories: mobility data computation and annotation. ACM Trans Intell Syst Technol 4(3):1–38
Yang C, Gidófalvi G (2020) Detecting regional dominant movement patterns in trajectory data with a convolutional neural network. Int J Geogr Inf Sci 34(5):996–1021
Yang Y, Li L, Zhang L et al (2022) Analysis of moving cluster with scene constraints for group behavior pattern mining. Neural Comput Appl 34(21):18545–18560
Yao R, Wang F, Chen S et al (2020) Groupseeker: an applicable framework for travel companion discovery from vast trajectory data. ISPRS Int J Geo-Inf 9(6):404
Yeoman J, Duckham M (2016) Decentralized detection and monitoring of convoy patterns. Int J Geogr Inf Sci 30(5):993–1011
Yu W (2018) Discovering frequent movement paths from taxi trajectory data using spatially embedded networks and association rules. IEEE Trans Intell Transp Syst 20(3):855–866
Yu W, Huang Q (2022) A deep encoder-decoder network for anomaly detection in driving trajectory behavior under spatio-temporal context. Int J Appl Earth Obs Geoinf 115:103115
Yuan G, Sun P, Zhao J et al (2017) A review of moving object trajectory clustering algorithms. Artif Intell Rev 47:123–144
Yuan H, Qian Y, Yang R et al (2014) Human mobility discovering and movement intention detection with GPS trajectories. Decis Support Syst 63:39–51
Yuan J, Zheng Y, Zhang C et al (2010) T-drive: driving directions based on taxi trajectories. Proceedings of the 18th SIGSPATIAL International Conference on Advances in Geographic Information Systems 99–108
Yuan Q, Shang J, Cao X et al (2017b) Detecting multiple periods and periodic patterns in event time sequences. Proceedings of the 2017 ACM on Conference on Information and Knowledge Management 617–626
Yuan Y, Wang G, Chen L et al (2016) Efficient pattern matching on big uncertain graphs. Inf Sci 339:369–394
Yue W, Li C, Chen Y et al (2021) What is the root cause of congestion in urban traffic networks: road infrastructure or signal control? IEEE Trans Intell Transp Syst 23(7):8662–8679
Zaki MJ (2001) Spade: an efficient algorithm for mining frequent sequences. Mach Learn 42:31–60
Zhang D, Lee K, Lee I (2018) Hierarchical trajectory clustering for spatio-temporal periodic pattern mining. Expert Syst Appl 92:1–11
Zhang J, Li J, Wang S et al (2014) On retrieving moving objects gathering patterns from trajectory data via spatio-temporal graph. 2014 IEEE International Congress on Big Data. IEEE 390–397
Zhang P, Van de Weghe N (2024) Discovering moving flock patterns in movement data: a reeb graph-based approach. Appl Sci 14(24):11883
Zhang X, Gao F, Wang J et al (2021) Evaluating a spatiotemporal shape-matching model for the generation of synthetic high spatiotemporal resolution time series of multiple satellite data. Int J Appl Earth Obs Geoinf 104:102545
Zhang Y, Ji G, Zhao B et al (2019) An algorithm for mining gradual moving object clusters pattern from trajectory streams. Comput Mater Continua 59(3):885–901
Zhang Z, Amiri H, Liu Z et al (2024b) Large language models for spatial trajectory patterns mining. Proceedings of the 1st ACM SIGSPATIAL International Workshop on Geospatial Anomaly Detection pp 52–55
Zhao B, Han J, Shi T et al (2020) Advancements in semantic trajectory modelling and mining. Journal of Geo-Information Science 22(4):842–856

Zhao B, Liu X, Jia J et al (2020b) A framework for group converging pattern mining using spatiotemporal trajectories. GeoInformatica 24:745–776
Zhao B, Liu M, Han J et al (2021) Efficient semantic enrichment process for spatiotemporal trajectories. Wirel Commun Mob Comput 2021(1):4488781
Zhao J, Itti L (2018) shapeDTW: shape dynamic time warping. Pattern Recogn 74:171–184
Zhao P, Liu X, Shi W et al (2020) An empirical study on the intraurban goods movement patterns using logistics big data. Int J Geogr Inf Sci 34(6):1089–1116
Zheng K, Zheng Y, Xie X et al (2012) Reducing uncertainty of low-sampling-rate trajectories. 2012 IEEE 28th international conference on data engineering. IEEE, pp 1144–1155
Zheng K, Zheng Y, Yuan NJ et al (2013) Online discovery of gathering patterns over trajectories. IEEE Trans Knowl Data Eng 26(8):1974–1988
Zheng Y (2015) Trajectory data mining: an overview. ACM Transactions on Intelligent Systems and Technology (TIST) 6(3):1–41
Zheng Y, Xie X, Ma WY (2010) GeoLife: a collaborative social networking service among user, location and trajectory. IEEE Data Eng Bull 33(2):32–39
Zimányi E, Sakr M, Lesuisse A (2020) Mobilitydb: a mobility database based on PostgreSQL and PostGIS. ACM Transactions on Database Systems (TODS) 45(4):1–42
Zhong L, Sun S, Law R (2019) Movement patterns of tourists. Tour Manage 75:318–322
Zhou X, Tan Y, Yu Z et al (2025) A method for mining spatial co-location patterns based on contextual similarity among categories. Journal of Geovisualization and Spatial Analysis 9(1):9
Zhou Z, Lin Y, Wen H et al (2024) PLM4Traj: cognizing movement patterns and travel purposes from trajectories with pre-trained language models. arXiv preprint arXiv:2405.12459. https://doi.org/10.48550/arXiv.2405.12459
Zhu Z, Xue Y, Chen X et al (2023) Large language models can learn rules. arXiv preprint arXiv:2310.07064. https://doi.org/10.48550/arXiv.2310.07064